

EE4580 - Quasi Optical Systems

Lecture # 2

Reflector Antennas

Nuria Llombart Juan

n.llombartjuan@tudelft.nl

LB.01.600, EWI

Topic #2 - Reflector Antennas

Covered sub-topics:

- Recap of the Equivalence Theorem
- Application of the Equivalence Theorem to the Analysis of Radiating Currents Distributions and Reflectors
- Radiation of a Uniform Circular Current Distribution (Airy Pattern)
- Analysis of Parabolic Reflector Antennas by the Method of the Geometrical Optics Combined to the Aperture Distribution Integral (GO+AI)
- Reflector efficiencies (in particular Taper Efficiency, Spillover Efficiency, and Aperture Efficiency)

Learning Objectives:

- Knowledge of the dominant physical phenomena involved in the radiation of currents distributions and reflectors
- Learning how to analyse reflector antennas by using the GO+AI approach
- Learning how to quantify the efficiencies of reflector antennas

Reflector Antennas – Applications

Arecibo Observatory



**Astronomical
Observations**

ALMA Radio-Telescope



**Cassini-Huygens
Mission**

Reflector Antennas – Applications



Airport surveillance radar antenna

RADAR – Civilian and Military Air Traffic Surveillance



Air traffic control radar antenna

Reflector Antennas – Applications

Telecommunication Systems

**Typical reflector antenna systems
for broadcasting communications**



**Parabolic antenna for satellite
communications at
Erdfunkstelle Raisting,
Germany**



Reflector Antennas – Applications

Civilian and Military Security Imaging Systems



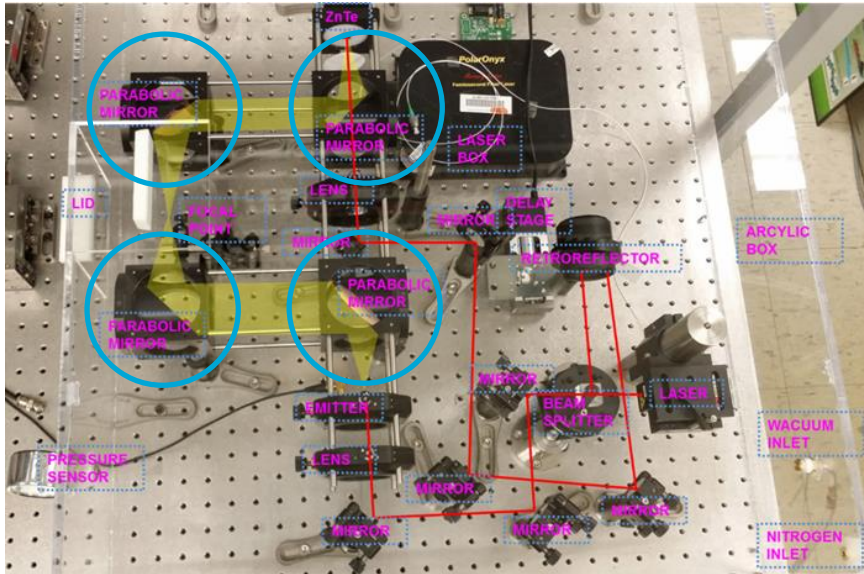
**K.B. Cooper, et al.,
2011**

Heinz, et al., 2014



Reflector Antennas – Applications

THz – Spectroscopy Systems

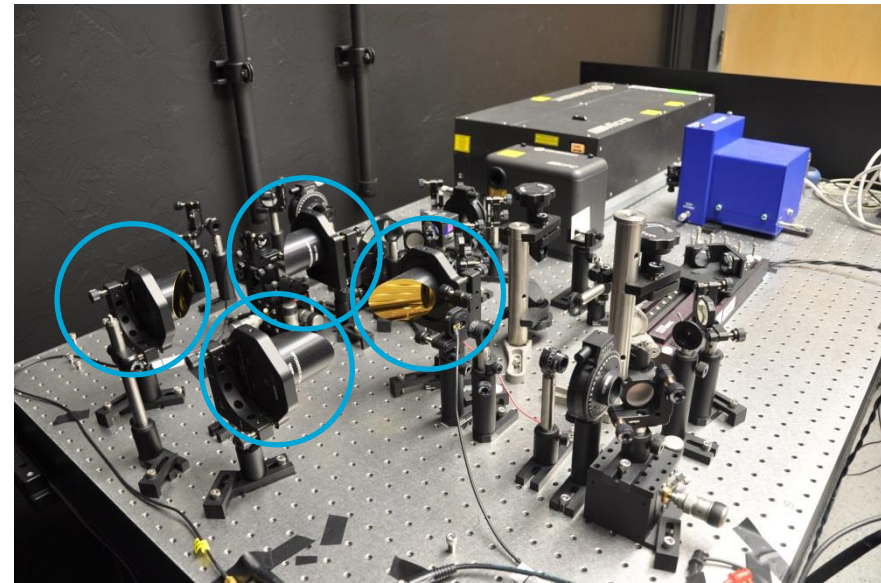


Florida International University

Typical Off-Axis
Parabolic Mirror
(Thorlabs)

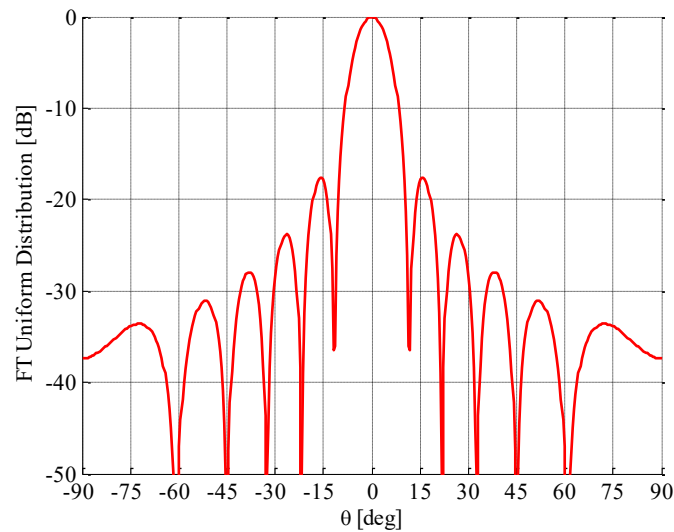
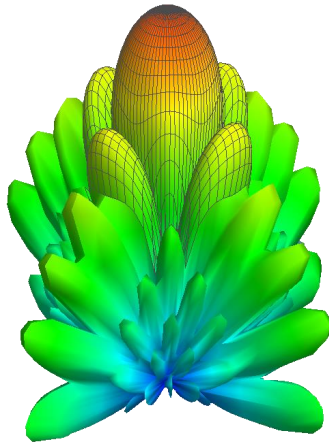


The University of Arizona

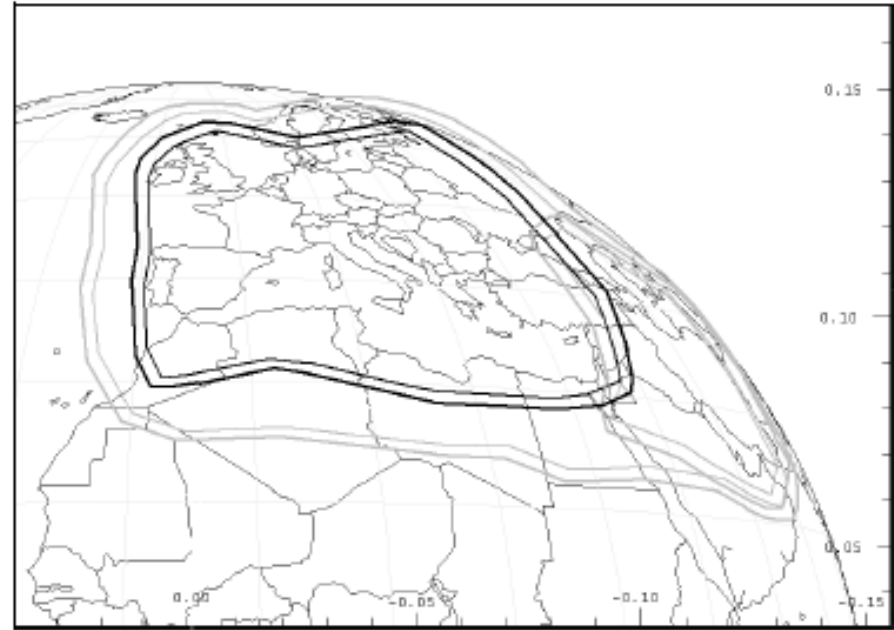


Why Reflector Antennas?

Highly Directive Beam



Shaped Beam

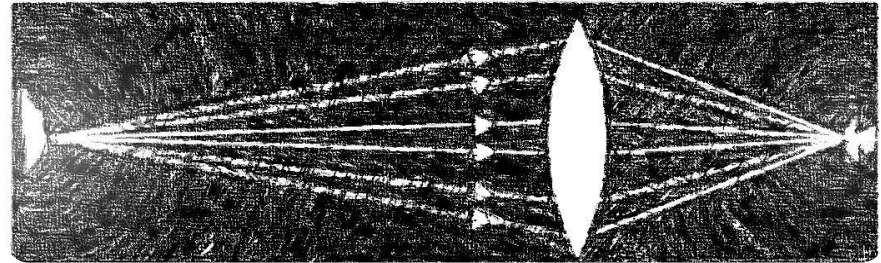


**Encinar, et al.,
2006**

Modelling a Quasi-Optical System

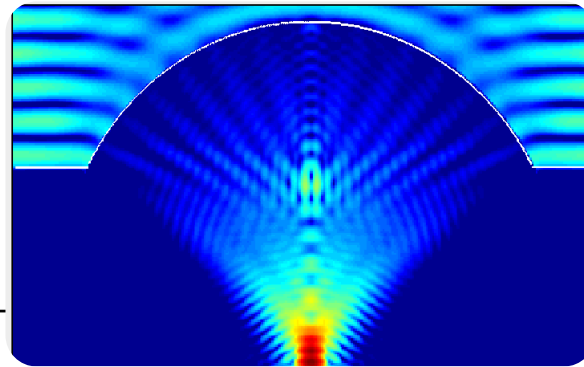
The EM field can be assumed, in first order, the superposition of the geometrical rays between the object and the system.

Geometrical Ray
Optics



Actual distribution is given by the Wave diffraction/interference at the optical system aperture which limits resolution

Wave
diffraction/interference



Parabolic Reflector

Geometrical Optics Considerations



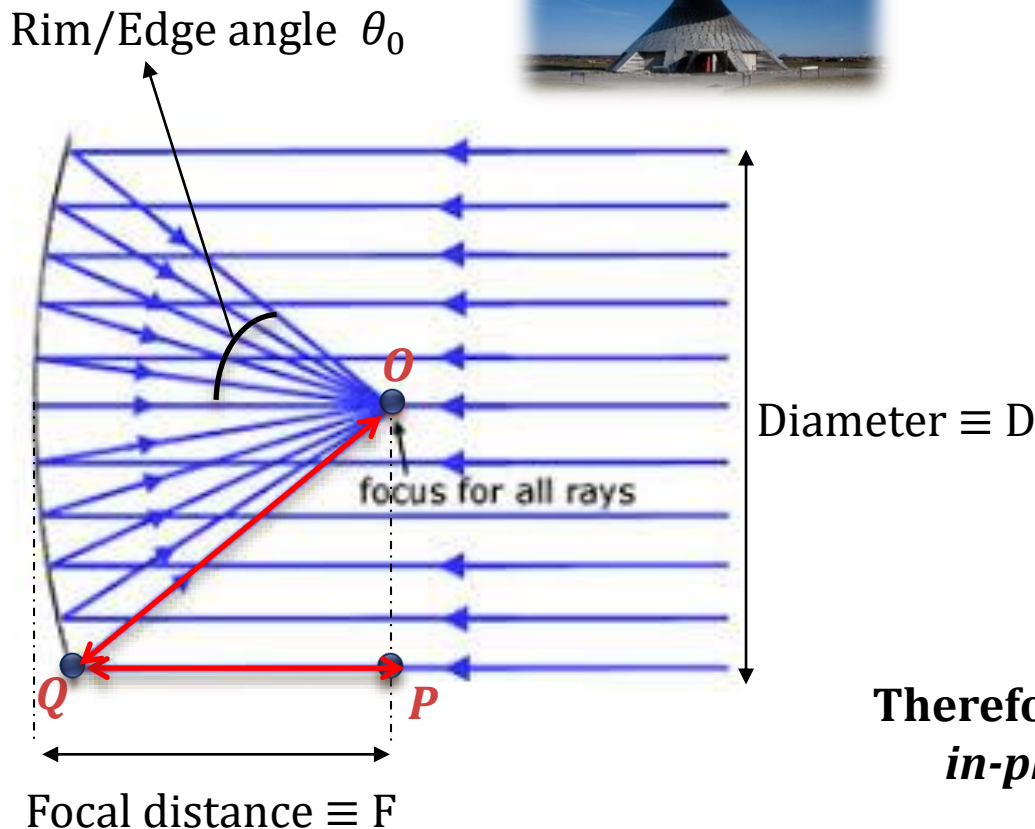
The parabolic surface ensures that all rays have the same length:

$$\text{Length: } L = |QO| + |PQ| = 2F$$

The path length relates to the phase of the EM field. Considering a plane wave model, the phase of a ray is given by

$$\text{Phase: } k_0 L = \frac{2\pi}{\lambda_0} L$$

Therefore, the rays reach the focus *in-phase* or at the *same time*

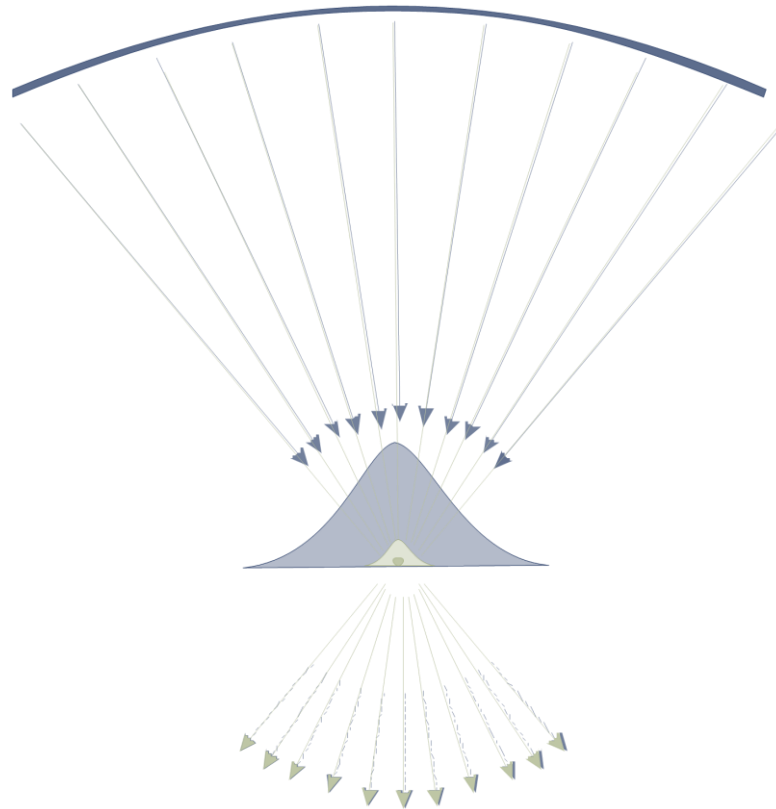
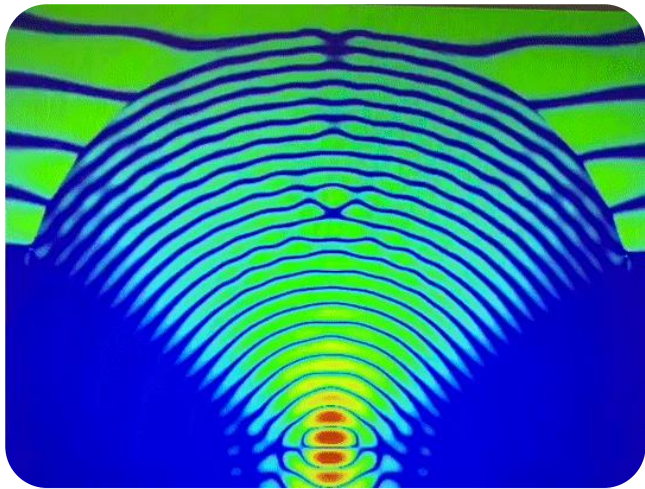


Field Distribution in the Focal Plane

Ray picture is independent of frequency but not field distribution

Red low frequency rays and distribution

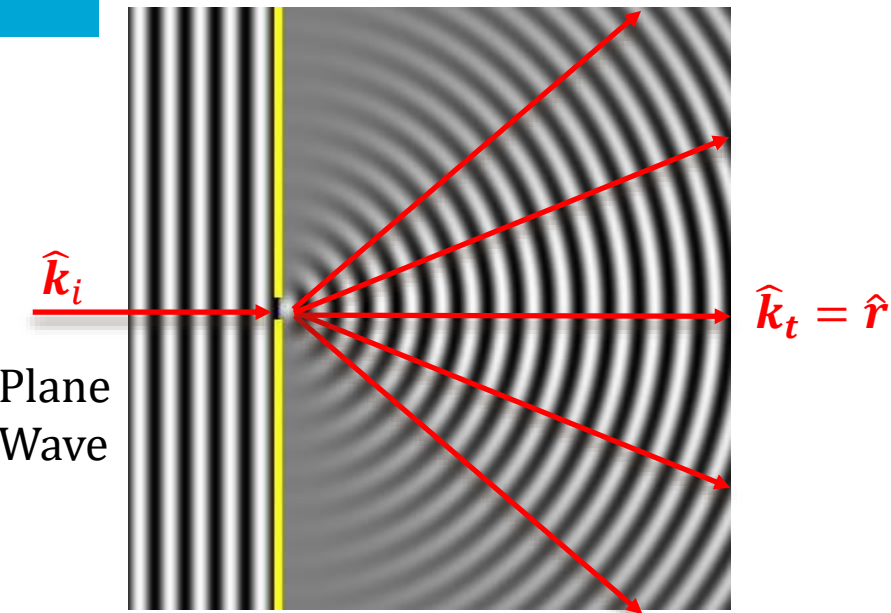
Blue high frequency rays and distribution



Drawing these pictures is intuitive
Quantifying them is not

Wave Diffraction and Interference

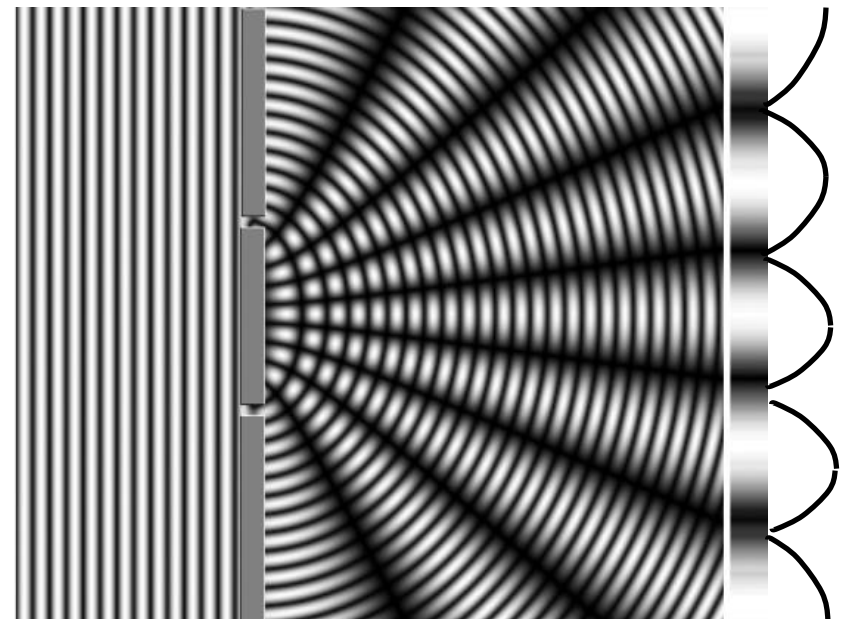
Metal sheet
with a small
opening



Diffracted wave
propagates in multiple
directions. It becomes a
spherical wave

Plane
Wave

Multiple
openings

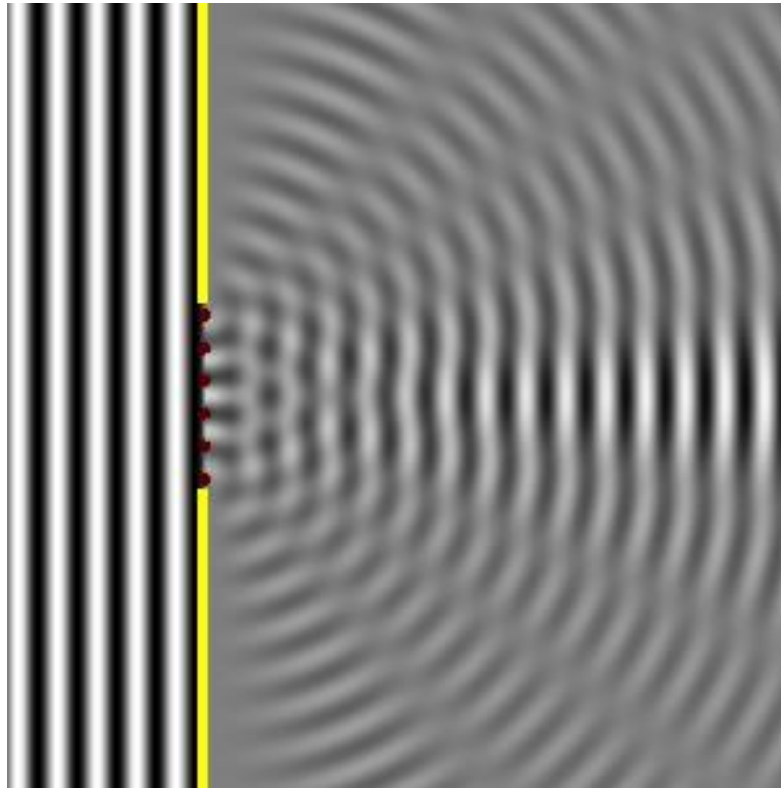


Constructive and
destructive interference
between the two
diffracted waves

Wave Diffraction and Interference

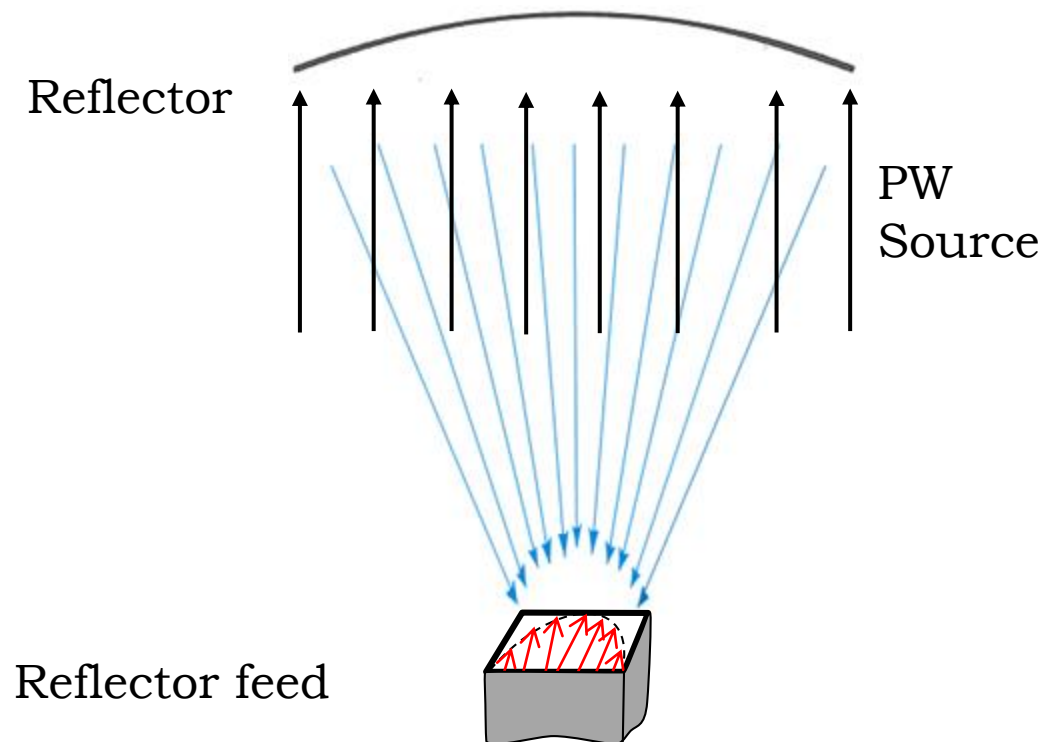
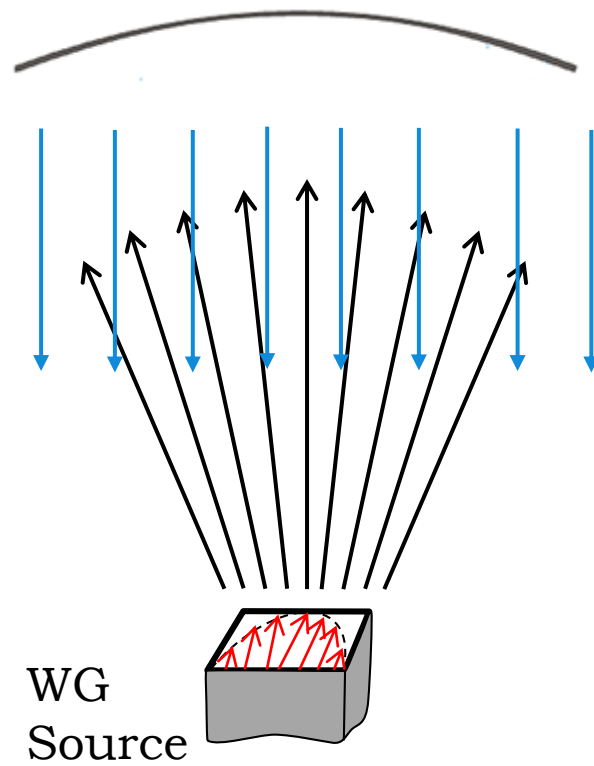
Plane
Wave

Metal sheet
with a large
opening



The two
effects mixed

Tx vs Rx Analysis

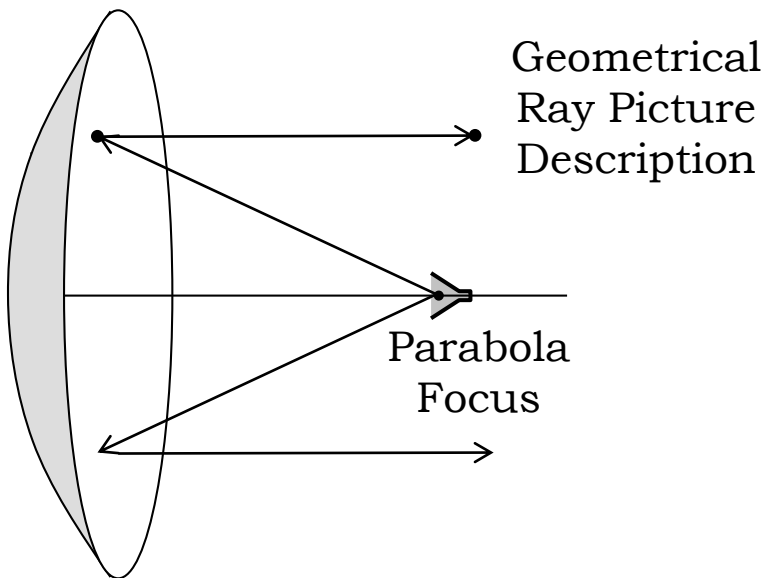


Reflector Antenna in Tx

Let us consider a typical reflector antenna:

it consists of a **parabolic shaped reflector** which is illuminated by a horn antenna placed at the focus of the parabola

Parabolic Reflector

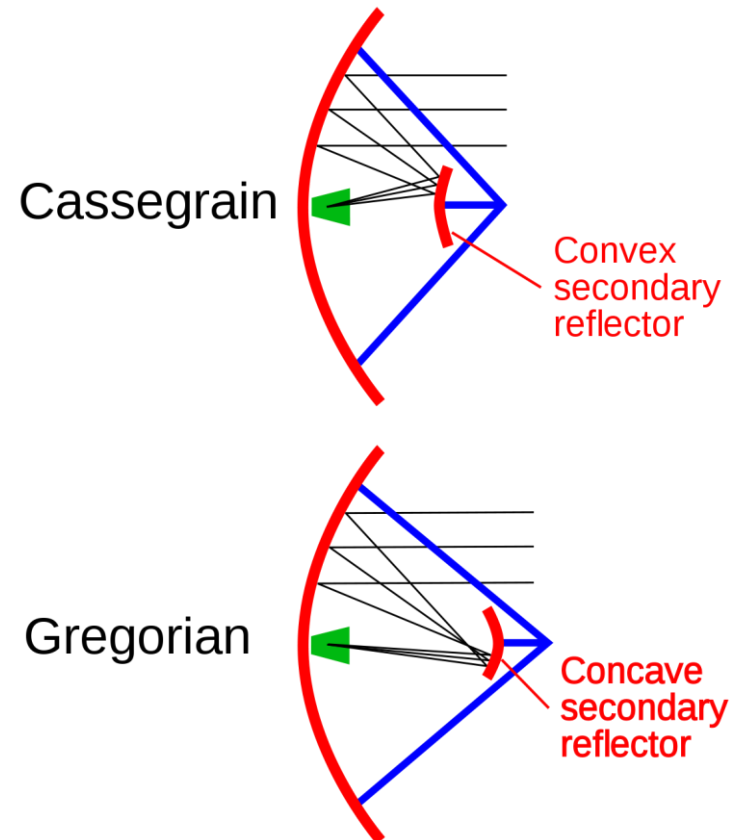
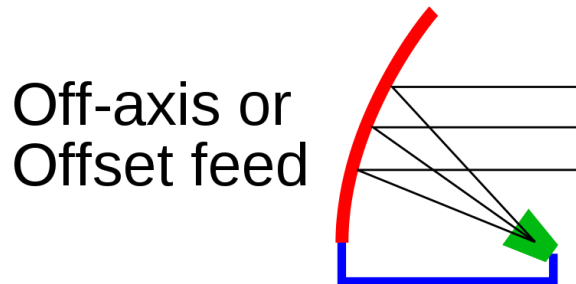
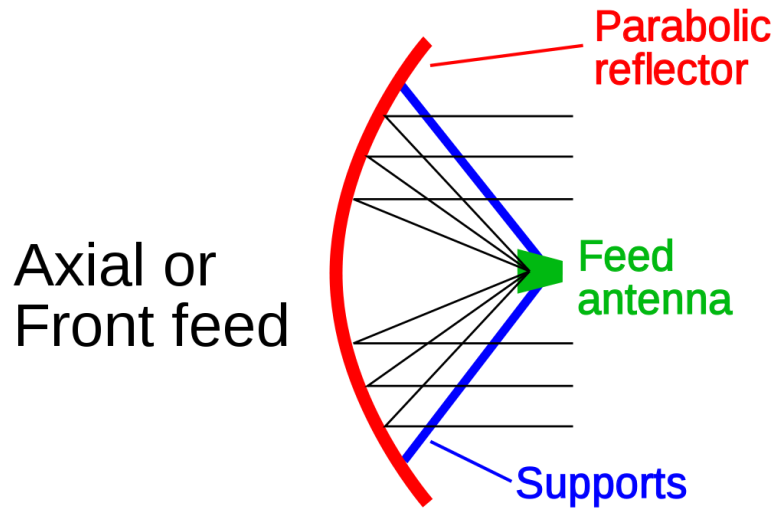


We would like to estimate the radiation pattern of the reflector

Realistic Hypotheses for the Analysis:

- the reflector lies in the far radiation zone of the feeder
- the reflector can be considered locally flat with respect to the wavelength
- the reflector is made of PEC and has a uniform reflection coefficient everywhere on the parabola

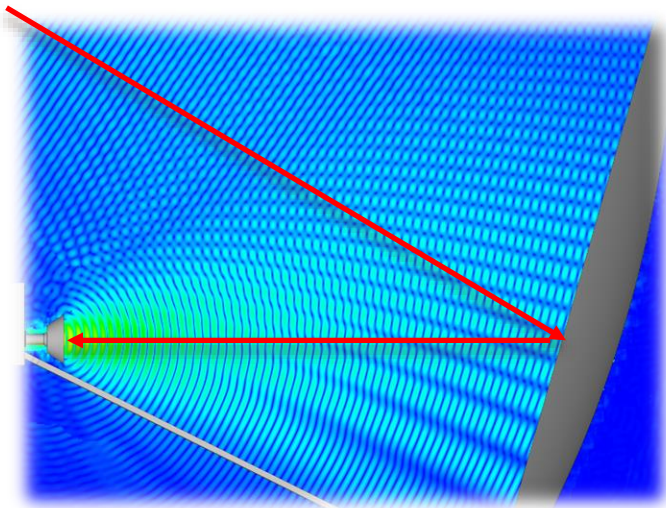
Practical Geometries vs Canonical Geometry



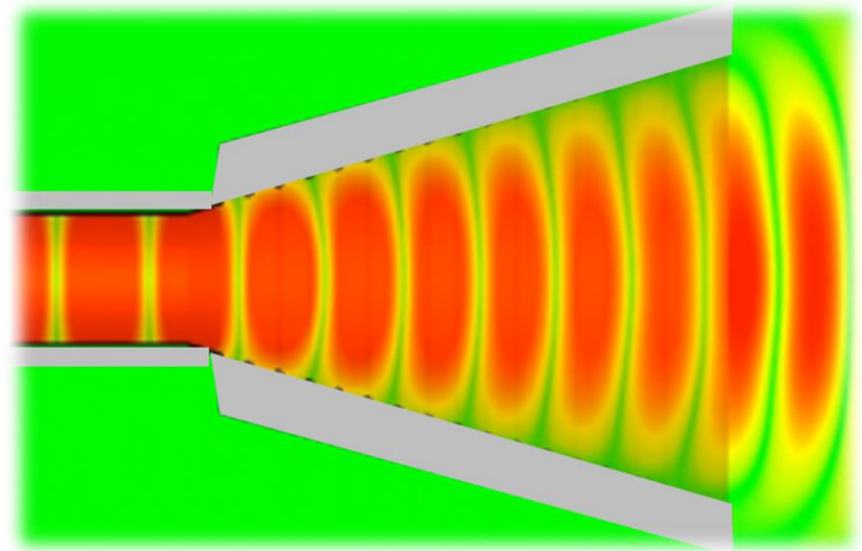
Wave Front Changes

*Tx: Spherical wave from the feed
to plane wave near the reflector aperture
to spherical wave in the far field*

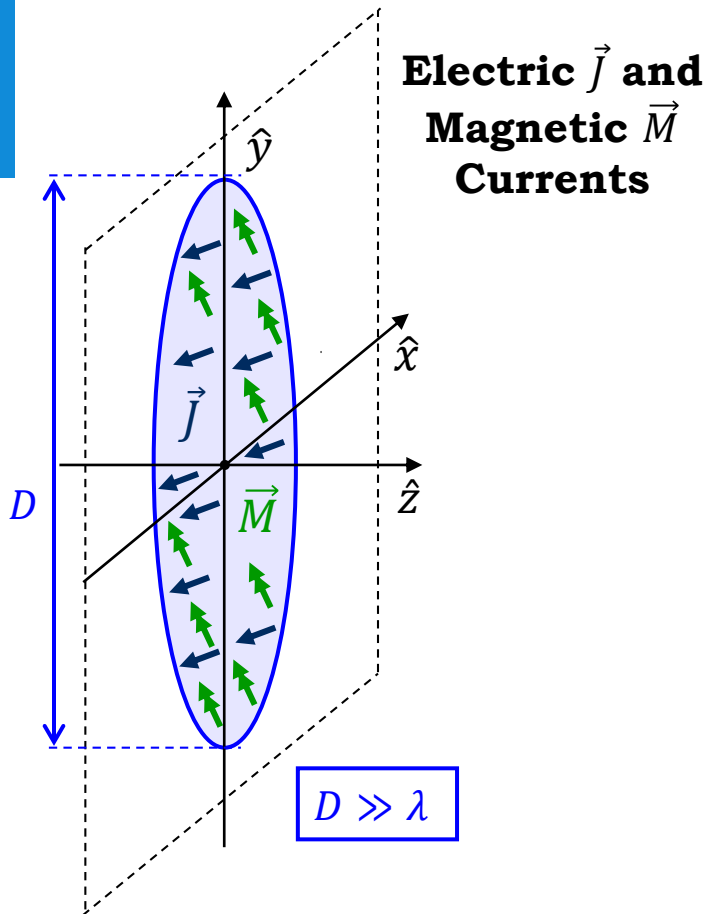
Reflector antenna



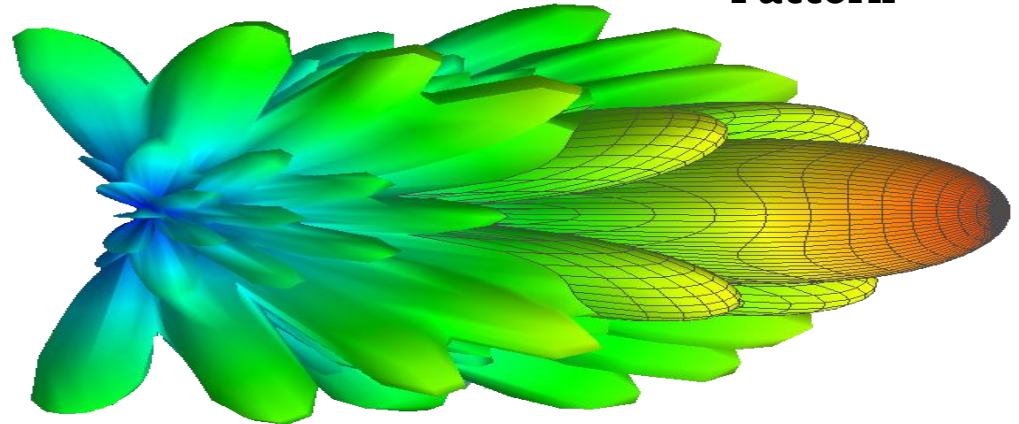
Reflector feed



Highly Directive Beams From Currents Distributions



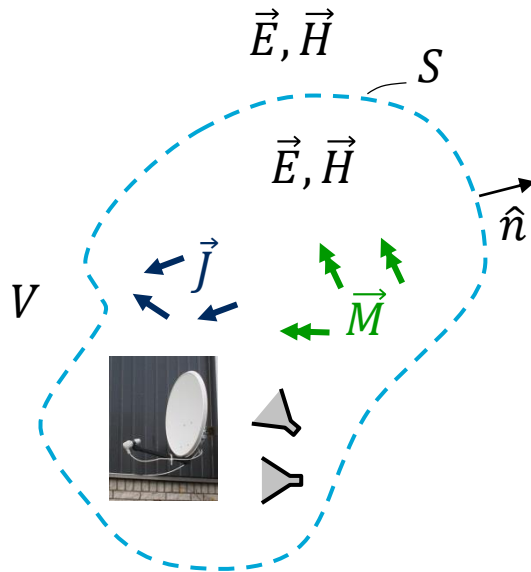
**Highly Directive
Far Field
Pattern**



The currents distributions can be real or equivalent in virtue of the Equivalence Theorem

Surface Equivalent Theorem – Recap (Love's Formulation)

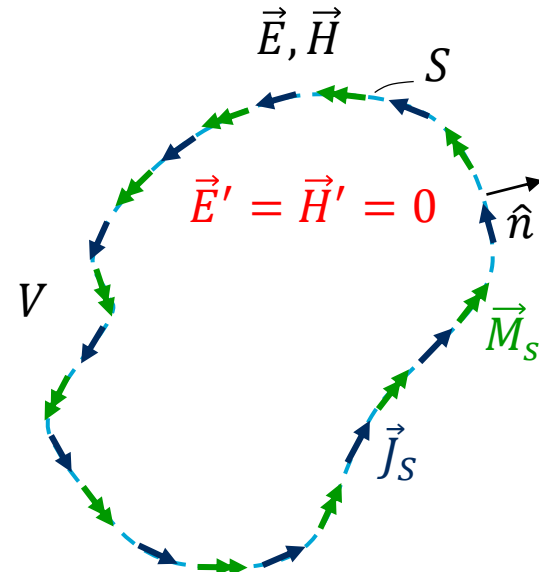
Original Problem



Magnetic and Electric
Equivalent Currents

$$\begin{aligned}\vec{M}_s &= \vec{E} \times \hat{n} \Big|_S \\ \vec{J}_s &= \hat{n} \times \vec{H} \Big|_S\end{aligned}$$

Equivalent Problem

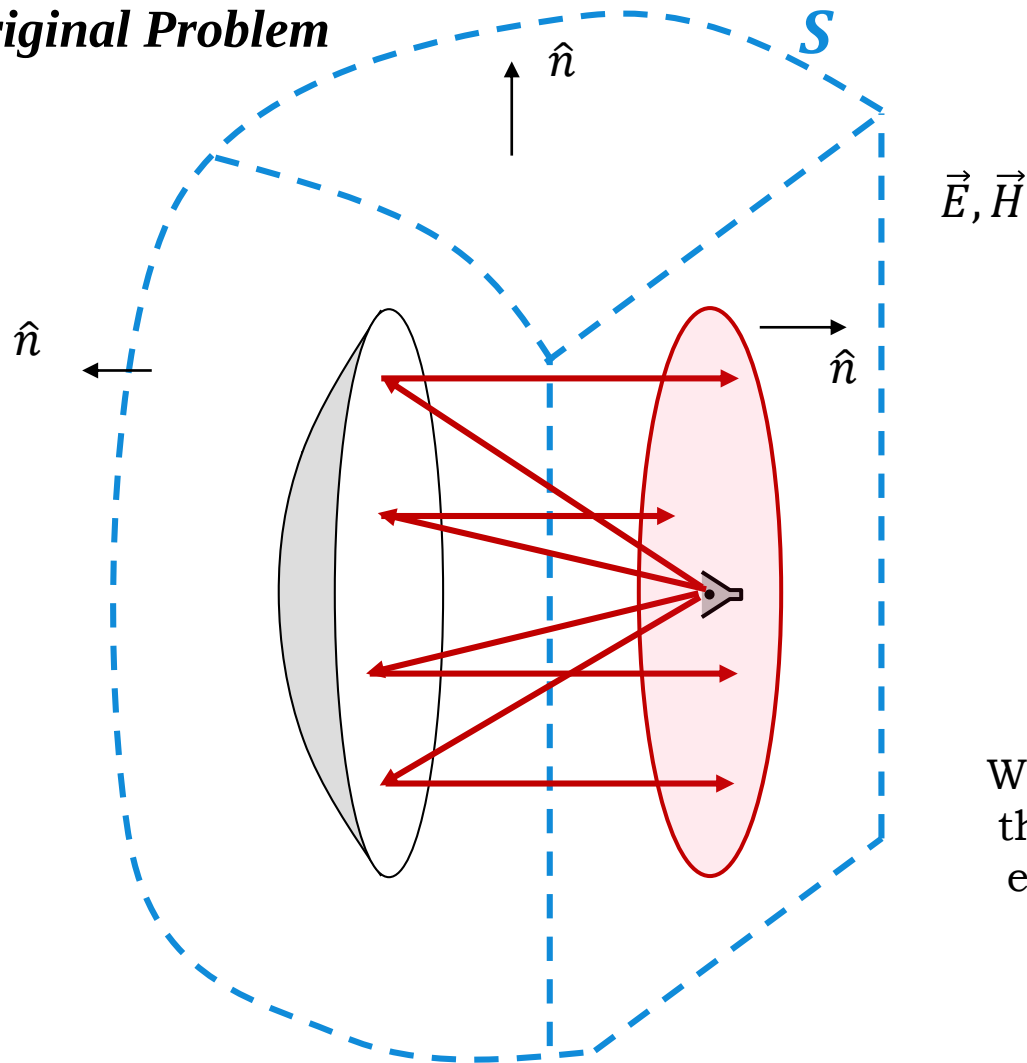


$\vec{E}, \vec{H}$ Fields of the
Original Problem on
the Surface

Roles of the Equivalent Currents: To satisfy the *Uniqueness Theorem* outside the surface S ; and to compensate the discontinuity of the field across the surface S

Equivalent Surface Theorem

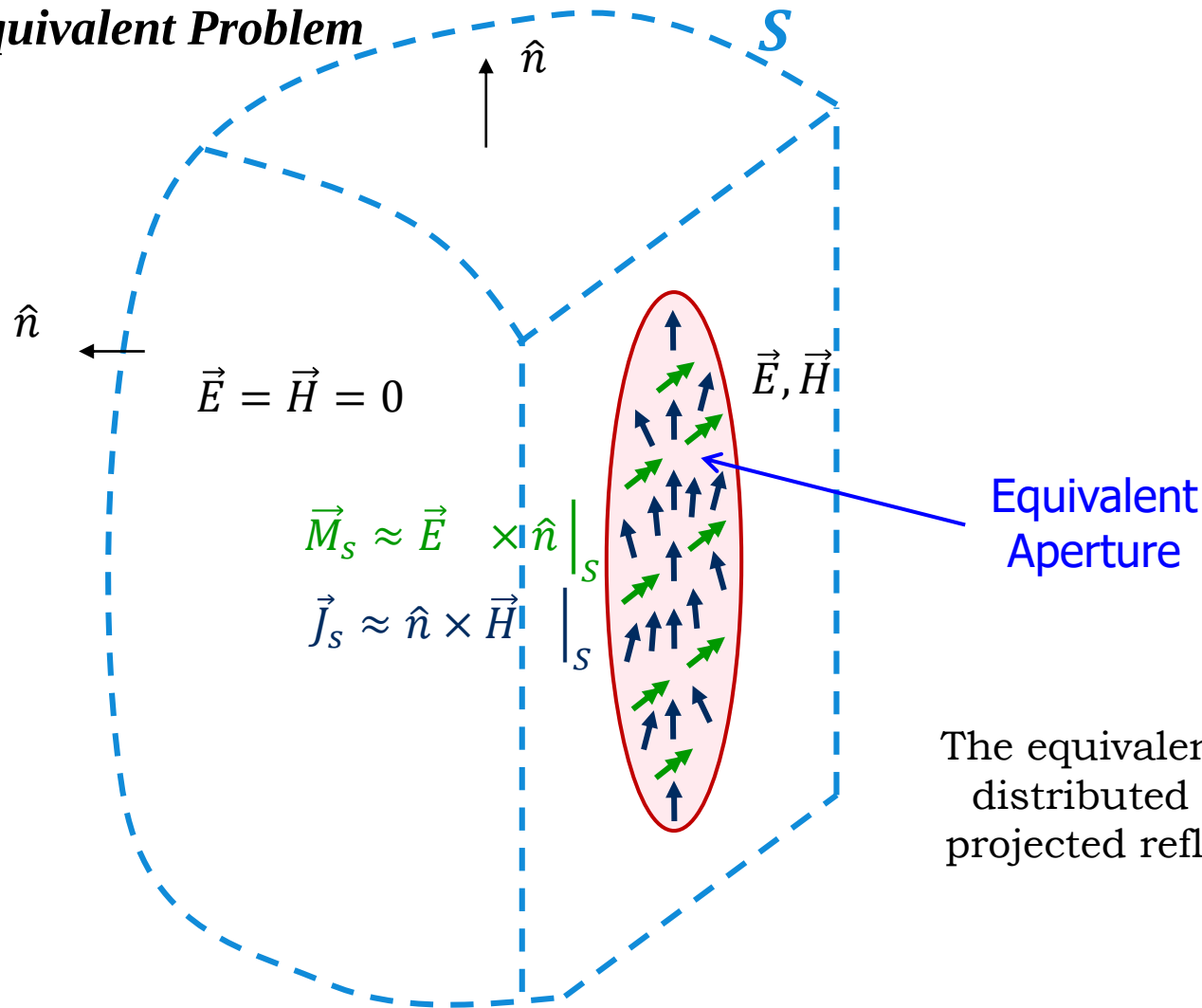
Original Problem



We simplify the calculation of the radiation by defining the equivalent currents only on the equivalent circular aperture

Equivalent Surface Theorem

Equivalent Problem



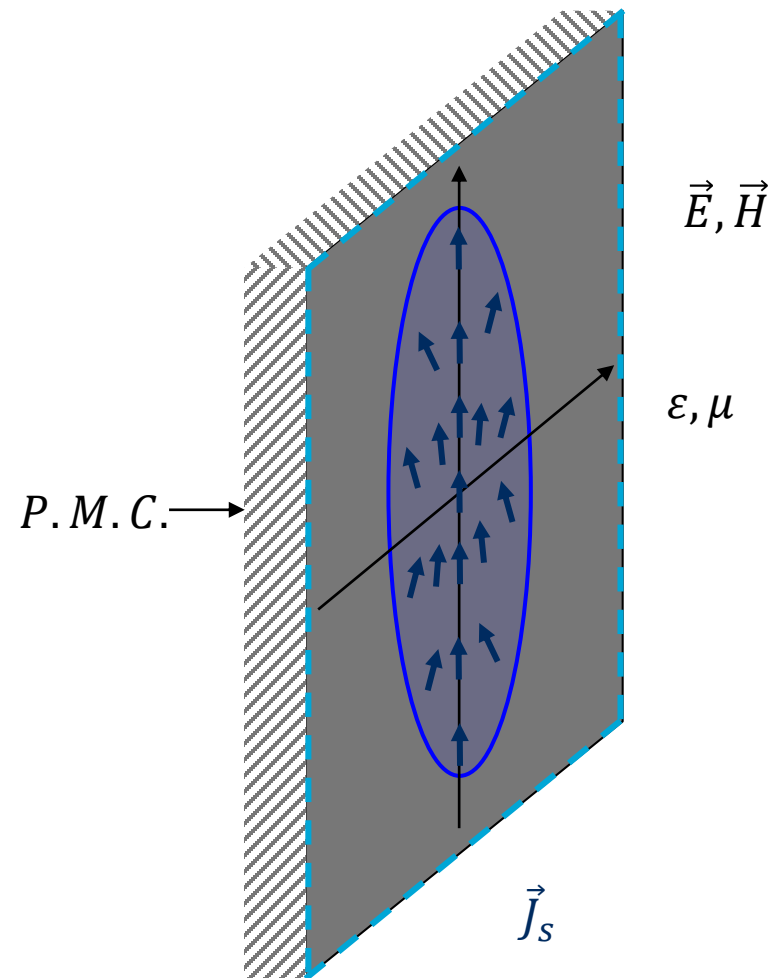
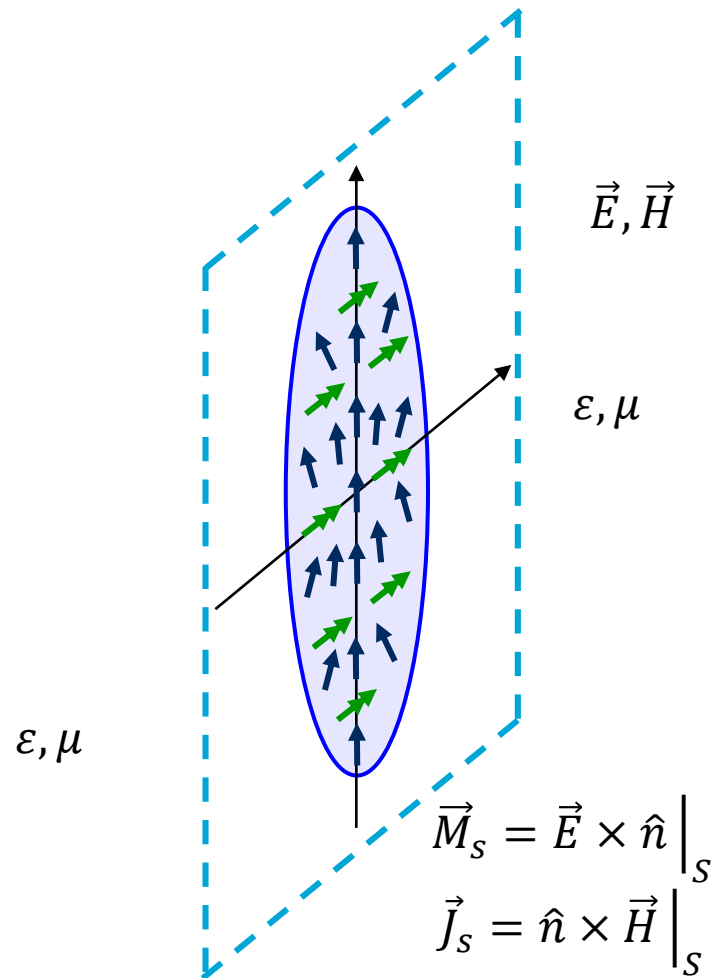
The equivalent currents are distributed only over the projected reflector aperture

Equivalent Aperture Integral Approach

Two choices

Love's formulation

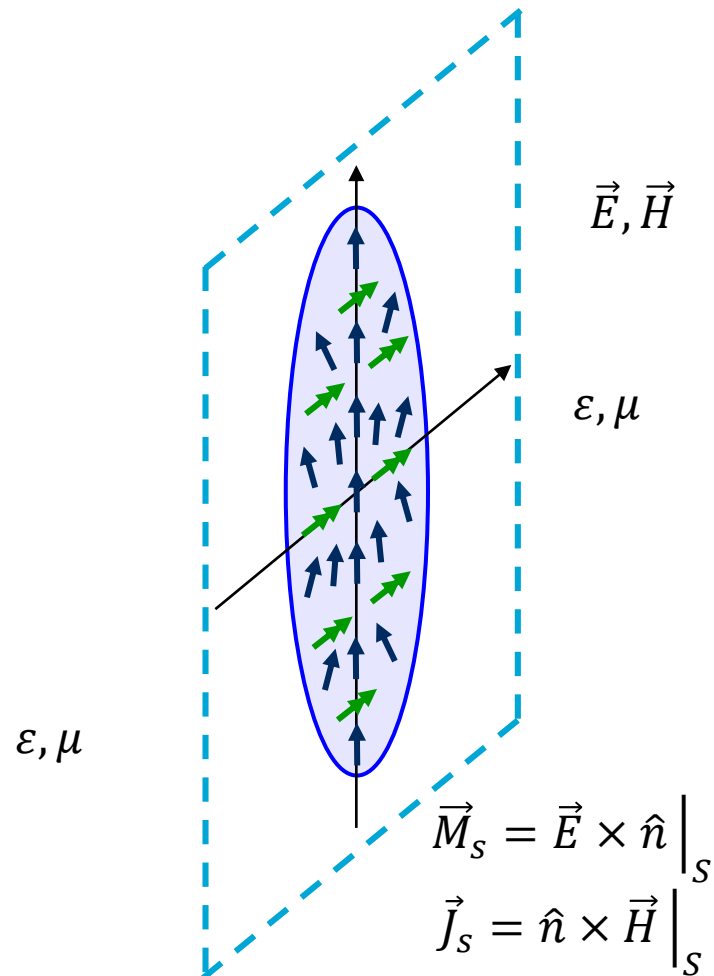
Schelkunoff's formulation



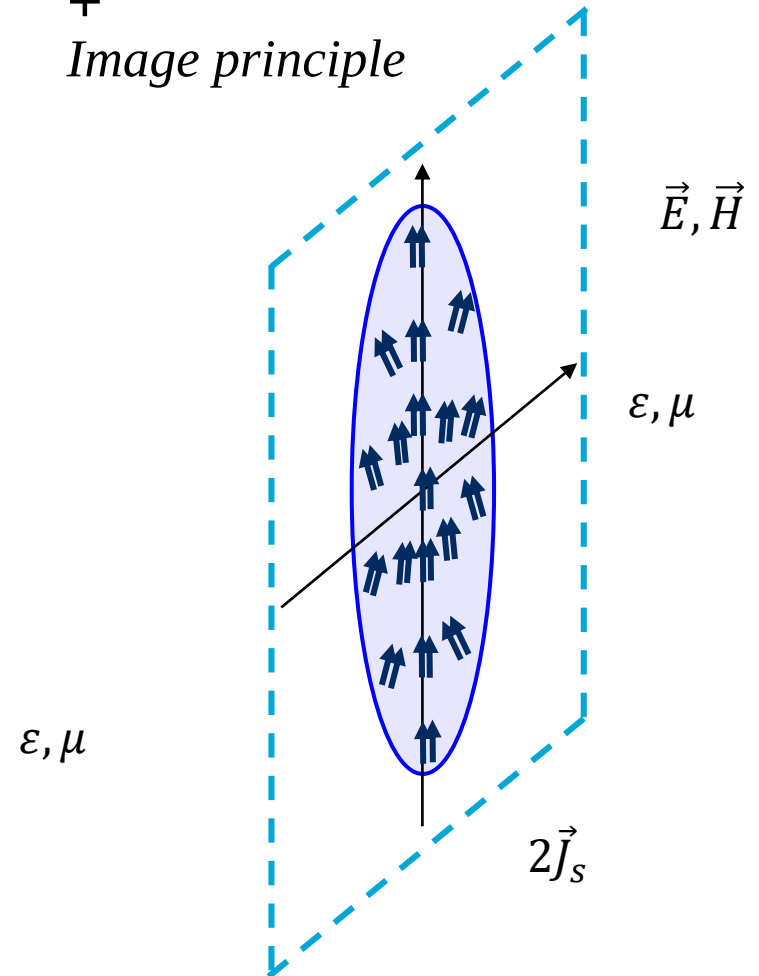
Equivalent Aperture Integral Approach

Two choices

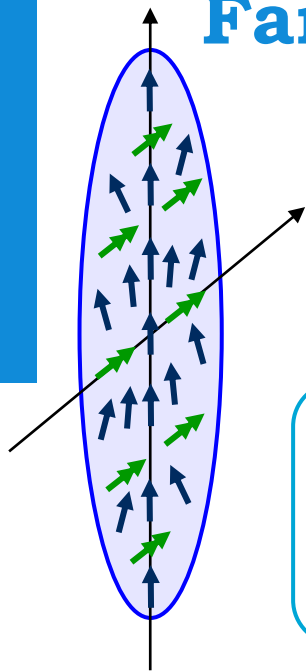
Love's formulation



*Schelkunoff's formulation
+
Image principle*



Far Field Estimation: Radiation of the Equivalent Currents



$$\begin{aligned}\vec{M} &= \vec{E} \times \hat{n} \Big|_S \\ \vec{J} &= \hat{n} \times \vec{H} \Big|_S\end{aligned}$$

Far Field proportional to
the Fourier Transform of
Currents Distribution

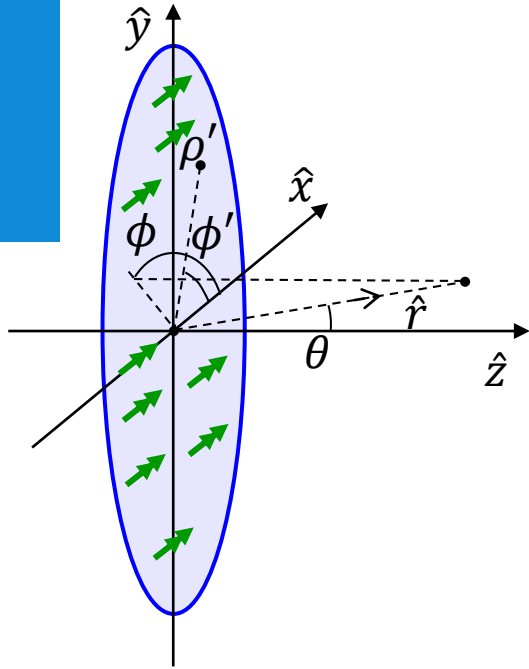
From Previous Lectures/EM Courses:
Far Field Estimation – Spectral Approach

$$\vec{f}^{far}(\vec{r}) \approx jk_{zs} \tilde{\mathbf{G}}^{fc}(k_{xs}, k_{ys}, z, z') \vec{\mathcal{C}}(k_{xs}, k_{ys}) \frac{e^{-jkr}}{2\pi r}$$

From Previous Lectures/EM Courses:
Far Field Estimation – Spatial Approach

$$\begin{aligned}\vec{E}^{far}(\vec{r}) &\approx \left[-\omega\mu(\tilde{I} - \hat{r}\hat{r}) \iint_S \vec{J}(\vec{r}') e^{jk\hat{r}\cdot\vec{r}'} dS + k\hat{r} \times \tilde{I} \iint_S \vec{M}(\vec{r}') e^{jk\hat{r}\cdot\vec{r}'} dS \right] \frac{je^{-jkr}}{4\pi r} \\ \vec{H}^{far}(\vec{r}) &\approx \left[-\omega\varepsilon(\tilde{I} - \hat{r}\hat{r}) \iint_S \vec{M}(\vec{r}') e^{jk\hat{r}\cdot\vec{r}'} dS - k\hat{r} \times \tilde{I} \iint_S \vec{J}(\vec{r}') e^{jk\hat{r}\cdot\vec{r}'} dS \right] \frac{je^{-jkr}}{4\pi r}\end{aligned}$$

“Far Field – Fourier Transform” of a Uniform Current Distribution on a Circular Domain



$$\vec{M}(k_x, k_y) = \iint_S \vec{M}(\vec{r}') e^{jk\hat{r} \cdot \vec{r}'} dS \quad a \text{ radius of the aperture}$$

$$\vec{M}(\vec{r}') = M_0 \hat{p}_m$$

Uniform Magnetic
Current Distribution
Oriented along $\hat{p}_m$

$$\vec{r}' = x' \hat{x} + y' \hat{y}$$

$$\hat{r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}$$

$$k_x = k \sin \theta \cos \phi$$

$$k_y = k \sin \theta \sin \phi$$

Cartesian/Spherical
Parametrization

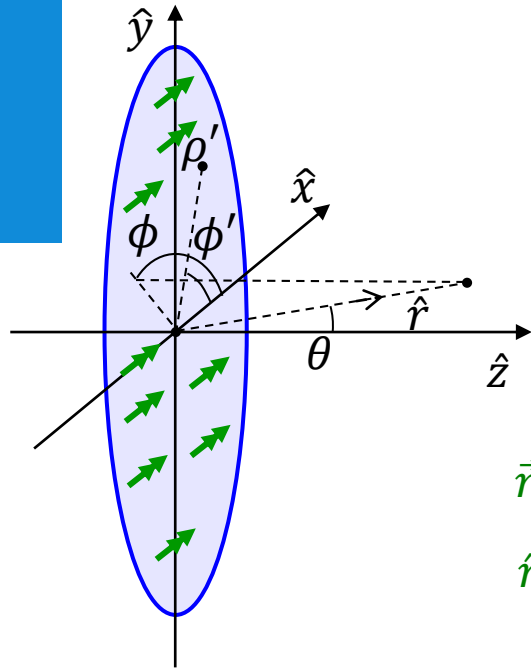
Phase Term in the Integral:

$$k\hat{r} \cdot \vec{r}' = kx' \sin \theta \cos \phi + ky' \sin \theta \sin \phi = k_x x' + k_y y'$$

$$\vec{M}(k_x, k_y) = \iint_S \vec{M}(\vec{r}') e^{jk\hat{r} \cdot \vec{r}'} dS = \iint_S \vec{M}(\vec{r}') e^{j(k_x x' + k_y y')} dS$$

Fourier Transform of
Currents Distribution

“Far Field – Fourier Transform” of a Uniform Current Distribution on a Circular Domain



$$\vec{M}(k_x, k_y) = \iint_S \vec{M}(\vec{r}') e^{jk\hat{r} \cdot \vec{r}'} dS \quad a \text{ radius of the aperture}$$

$$\vec{M}(\vec{r}') = M_0 \hat{p}_m$$

Uniform Magnetic
Current Distribution
Oriented along $\hat{p}_m$

$$\vec{r}' = \rho' \cos \phi' \hat{x} + \rho' \sin \phi' \hat{y}$$

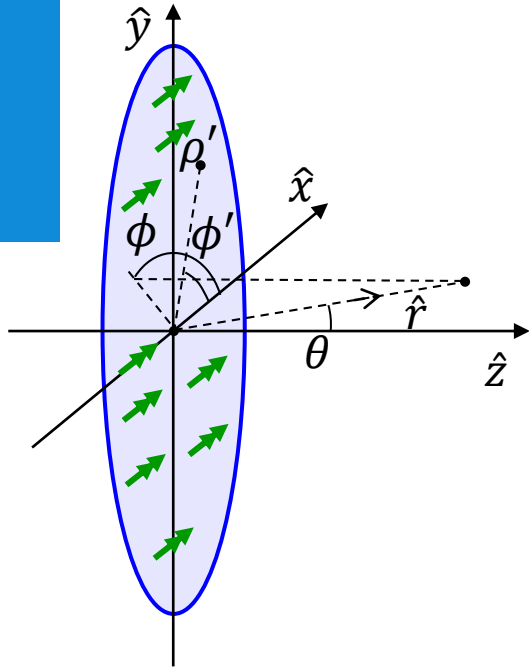
$$\hat{r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}$$

Polar/Spherical
Parametrization

Phase Term in the Integral:

$$\begin{aligned} k\hat{r} \cdot \vec{r}' &= k\rho' \cos \phi' \sin \theta \cos \phi + k\rho' \sin \phi' \sin \theta \sin \phi \\ &= k\rho' \sin \theta (\cos \phi' \cos \phi + \sin \phi' \sin \phi) \\ &= k\rho' \sin \theta \cos(\phi - \phi') \end{aligned}$$

“Far Field – Fourier Transform” of a Uniform Current Distribution on a Circular Domain



$$\begin{aligned}\vec{M}(k_x, k_y) &= \iint_S \vec{M}(\vec{r}') e^{jk\hat{r} \cdot \vec{r}'} dS \\ &= M_0 \hat{p}_m \int_0^a \int_0^{2\pi} e^{jk\rho' \sin \theta \cos(\phi - \phi')} \rho' d\rho' d\phi'\end{aligned}$$

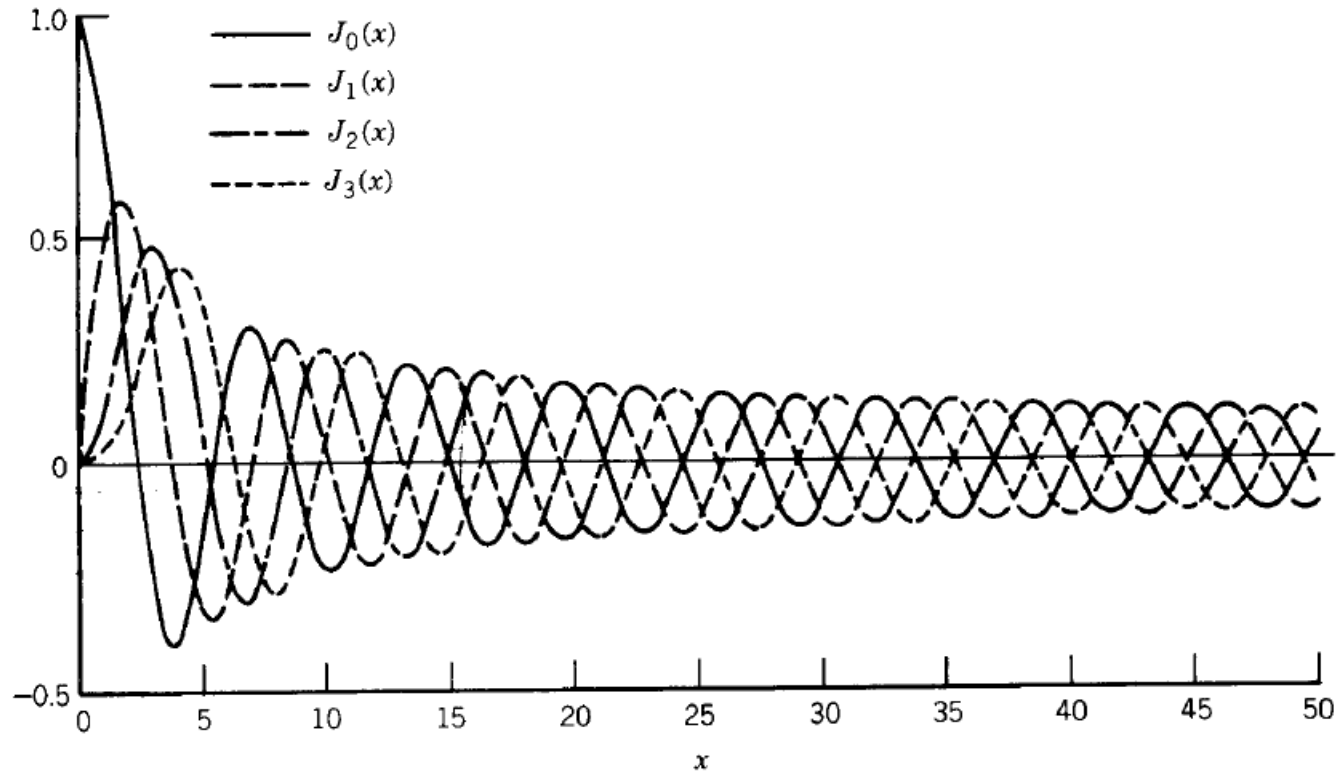
0-Order Bessel Function J_0 Identity

$$\int_0^{2\pi} e^{jk\rho' \sin \theta \cos(\phi - \phi')} d\phi' = 2\pi J_0(k\rho' \sin \theta)$$

$$\vec{M}(k_x, k_y) = M_0 \hat{p}_m 2\pi \int_0^a J_0(k\rho' \sin \theta) \rho' d\rho'$$

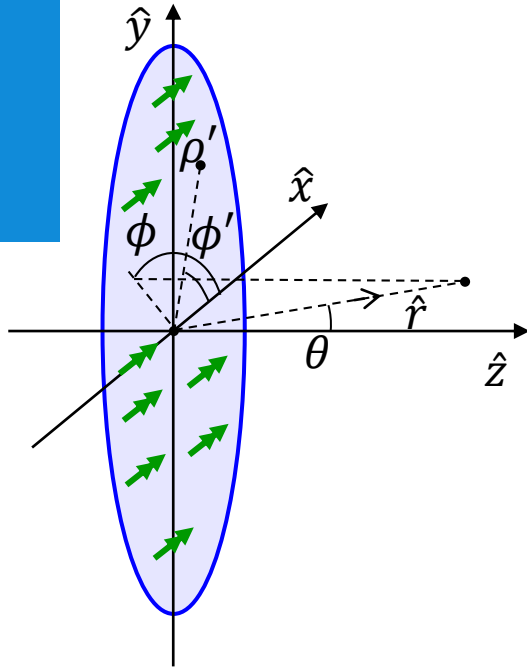
Bessel Functions

$$J_0(x), J_1(x), \dots, J_n(x)$$



Balanis, Antenna Theory

“Far Field – Fourier Transform” of a Uniform Current Distribution on a Circular Domain



$$\vec{M}(k_x, k_y) = M_0 \hat{p}_m 2\pi \int_0^a J_0(k\rho' \sin \theta) \rho' d\rho'$$

Bessel Functions Integral Property

$$\int t J_0(t) dt = t J_1(t)$$

$$\vec{M}(k_x, k_y) = M_0 \hat{p}_m 2\pi \frac{1}{(k \sin \theta)^2} [t J_1(t)]_{t=0}^{t=ka \sin \theta}$$

$$\vec{M}(k_x, k_y) = M_0 \hat{p}_m 2\pi \frac{ka \sin \theta}{(k \sin \theta)^2} J_1(ka \sin \theta)$$

$$\vec{M}(k_x, k_y) = 2M_0 \hat{p}_m \frac{\pi a^2 J_1(ka \sin \theta)}{ka \sin \theta}$$

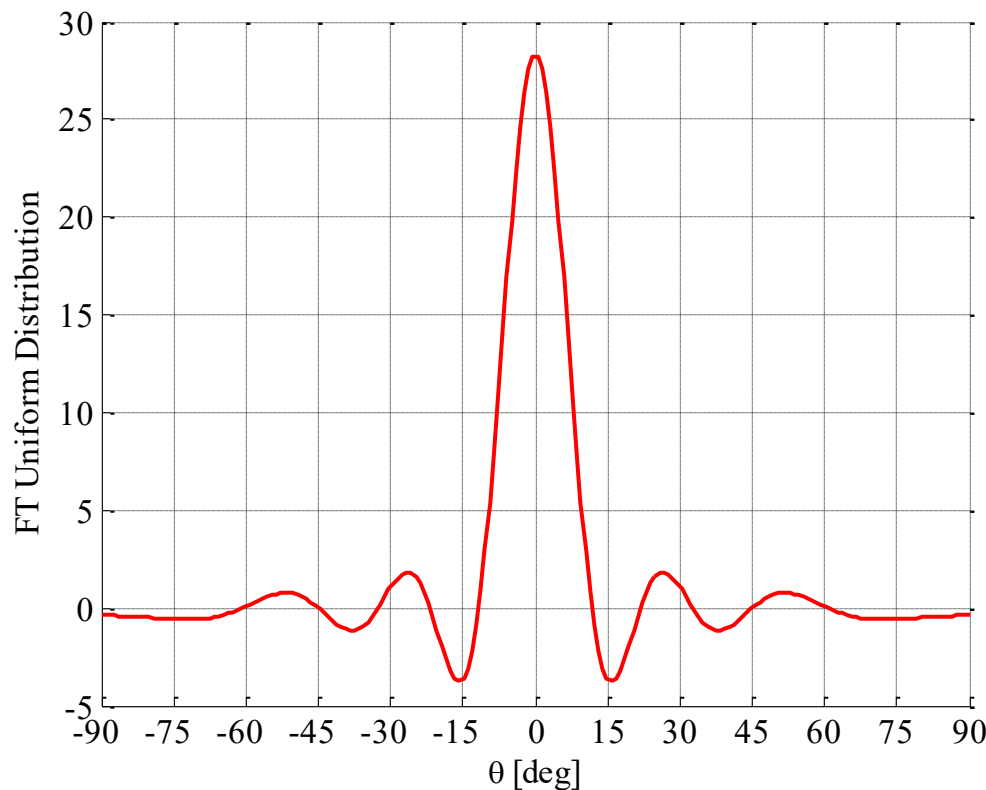
$$\vec{M}(k_x, k_y) = M_0 \hat{p}_m 2A \frac{J_1(ka \sin \theta)}{ka \sin \theta}$$

Airy Pattern

“Far Field – Fourier Transform” of a Uniform Current Distribution on a Circular Domain

$$\vec{M}(k_x, k_y) = M_0 \hat{p}_m 2A \frac{J_1(ka \sin \theta)}{ka \sin \theta} \quad \text{Does not depend on } \phi; \text{ cylindrically symmetric!}$$

Airy Pattern



$$a = 3\lambda, \lambda = 1\text{m}$$
$$A = \pi a^2$$

$$D = \frac{4\pi}{\lambda^2} A$$
$$= \frac{4\pi}{\lambda^2} (\pi a^2) = \left(\frac{2\pi a}{\lambda} \right)^2$$

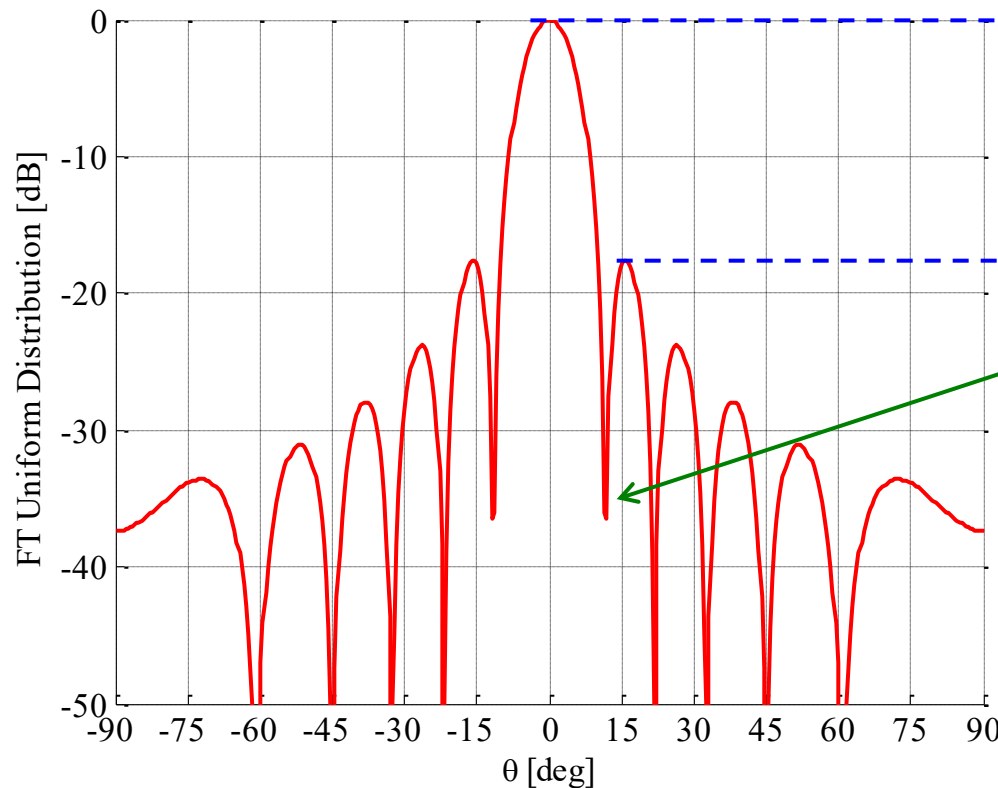
“Far Field – Fourier Transform” of a Uniform Current Distribution on a Circular Domain

$$\vec{M}(k_x, k_y) = M_0 \hat{p}_m 2A \frac{J_1(ka \sin \theta)}{ka \sin \theta}$$

$$a = 3\lambda, \lambda = 1\text{m}, D = 2a$$

$$A = \pi a^2$$

Airy Pattern



$SLL \cong -17.6\text{dB}$
(Side Lobe Level)

First Null (rad) = $1.22 \frac{\lambda}{D}$

Directivity

$$D_{dB} = 10 \log_{10} \left(\frac{2\pi a}{\lambda} \right)^2$$

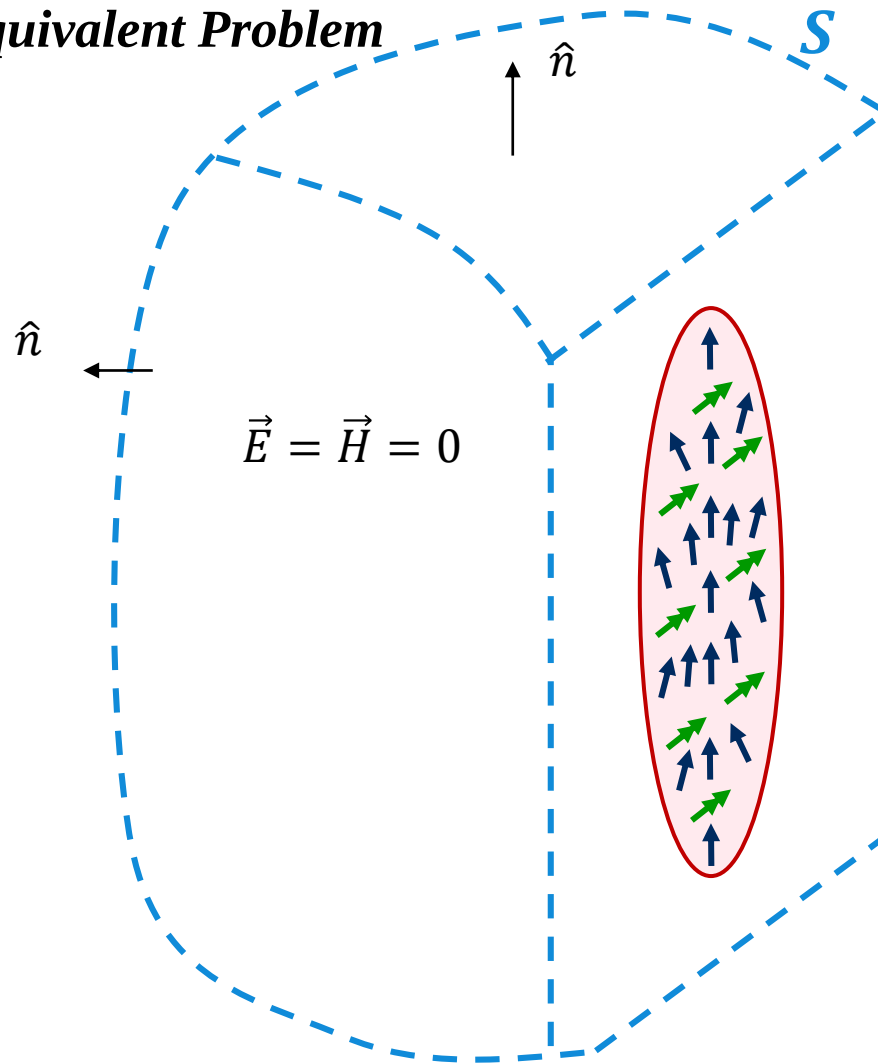
Half Power Beamwidth

$$\Delta\theta_{-3dB} = \lambda/D$$

Physical Optics Approximation

How can we derive an expression for the equivalent current distribution?

Equivalent Problem

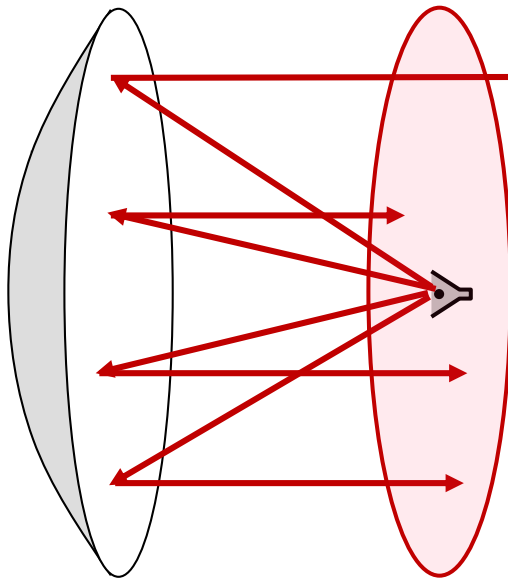


Physical Optics (PO)
Approximation:

1. The equivalent currents are obtained by using the incident field on the equivalent aperture
2. Outside this aperture, the equivalent currents are zero

Geometrical Ray Method

- The field incident over the equivalent aperture is the reflected by the reflector

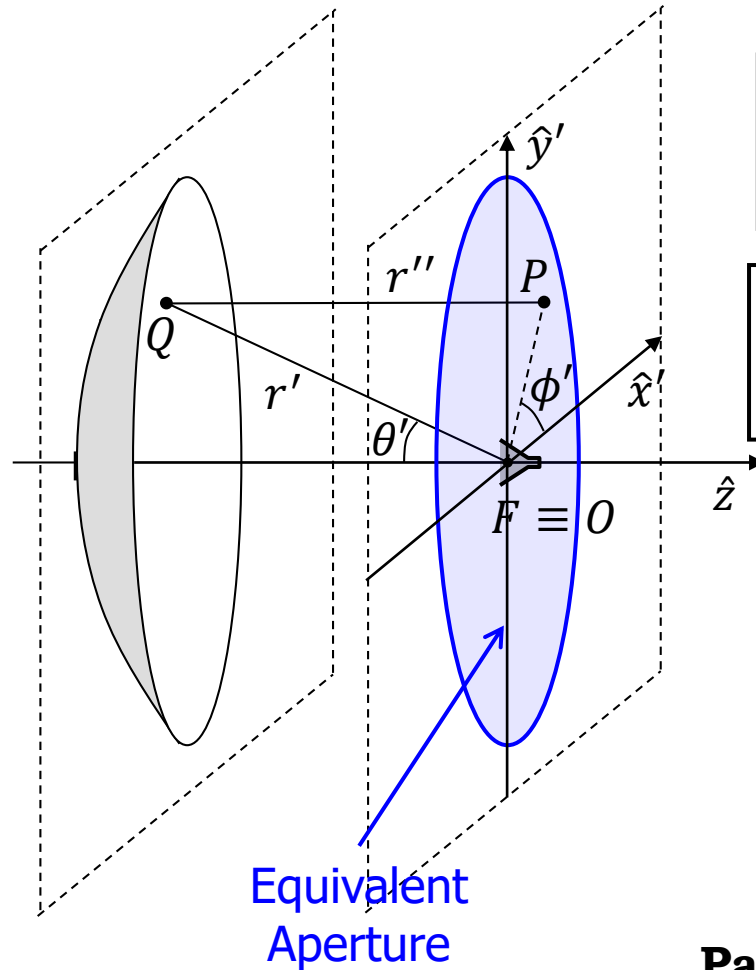


$$\begin{aligned}\vec{E}_a(\rho', \phi') &\approx \vec{E}_r(\rho', \phi') & \vec{M}_s &\approx \vec{E}_r \times \hat{n} \Big|_s \\ \vec{H}_a(\rho', \phi') &\approx \vec{H}_r(\rho', \phi') & \vec{J}_s &\approx \hat{n} \times \vec{H}_r \Big|_s\end{aligned}$$

- The reflected field can be estimated via Geometrical Optics (ray picture) where the surfaces are considered locally flat

$$\vec{E}_r(P) = \tilde{R}(Q) \cdot \vec{E}_i(Q) S(Q) e^{-jk_0 r''}$$

Parabolic Reflector Antennas



A spherical wave radiated from the feed is transformed in a transverse electromagnetic wave propagating along $\hat{z}$

At the aperture plane, all the field contributions have the same phase [$k_0(2f)$ with respect to the phase at the origin]

- Lengths of the rays FQ and QP:
 $|P - Q| = r'' \quad |Q - F| = r'$
- $r'' + r'$ total optical path length for every θ', ϕ'

Parabola Definition: $|P - Q| + |Q - F| = 2f$

f focal length

Geometrical Description & Parametrizations

$$|P - Q| + |Q - F| = 2f$$

$$|P - Q| = r' \cos \theta' = r''$$

$$|Q - F| = r'$$

From the definition: $r' \cos \theta' + r' = 2f$

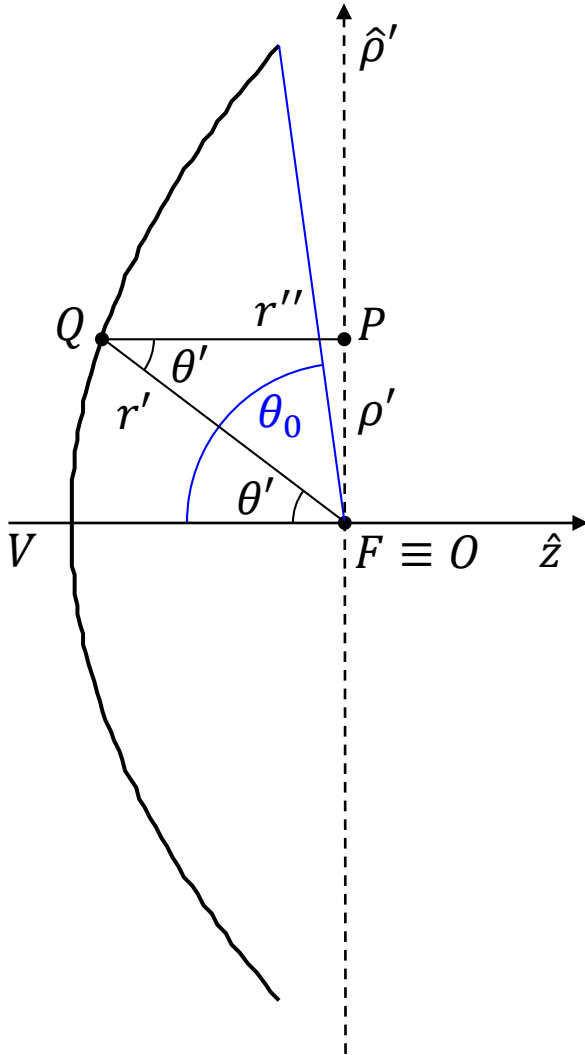
(r', θ') parametrization:

$$r' = \frac{2f}{1 + \cos \theta'} = \frac{f}{\cos^2 \left(\frac{\theta'}{2} \right)} = f \left[1 + \tan^2 \left(\frac{\theta'}{2} \right) \right]$$

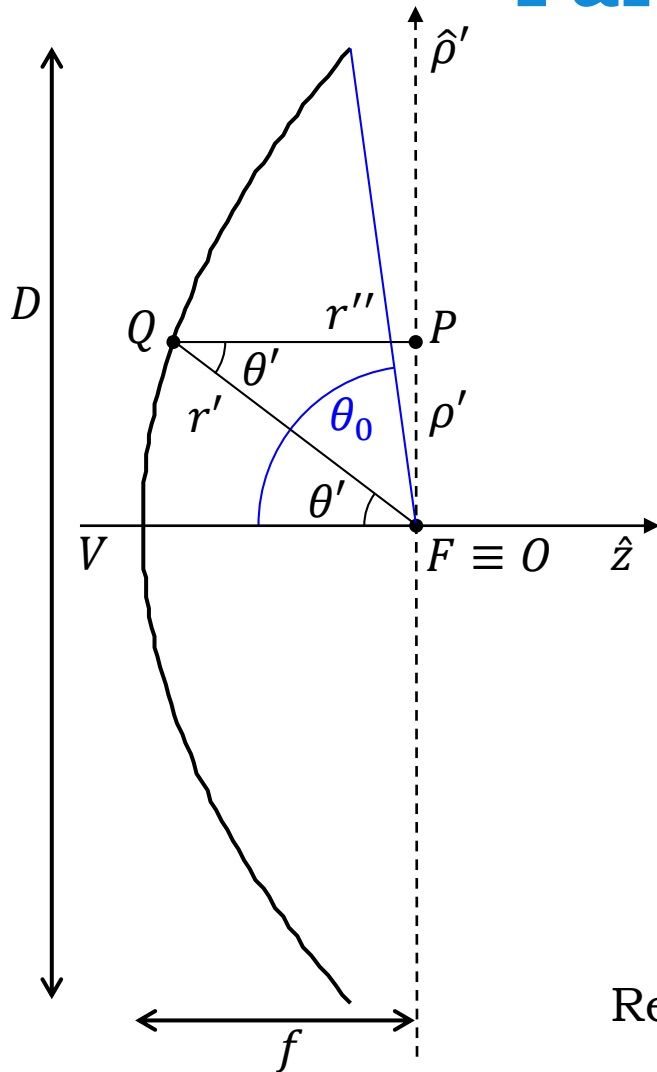
Alternative Parametrizations:

$$(r', \rho') \text{ parametrization: } r' = f \left[1 + \left(\frac{\rho'}{2f} \right)^2 \right]$$

$$(\rho', z) \text{ parametrization: } z = -f + \frac{(\rho')^2}{4f}$$



Geometrical Description & Parametrizations



(r', θ') parametrization:

$$r' = \frac{2f}{1 + \cos \theta'}$$

$$\rho' = r' \sin \theta' = \frac{2f \sin \theta'}{1 + \cos \theta'} = 2f \tan \left(\frac{\theta'}{2} \right)$$



The feed sees the edge of the reflector with an angle of θ_0
(reflector edge or subtended angle)

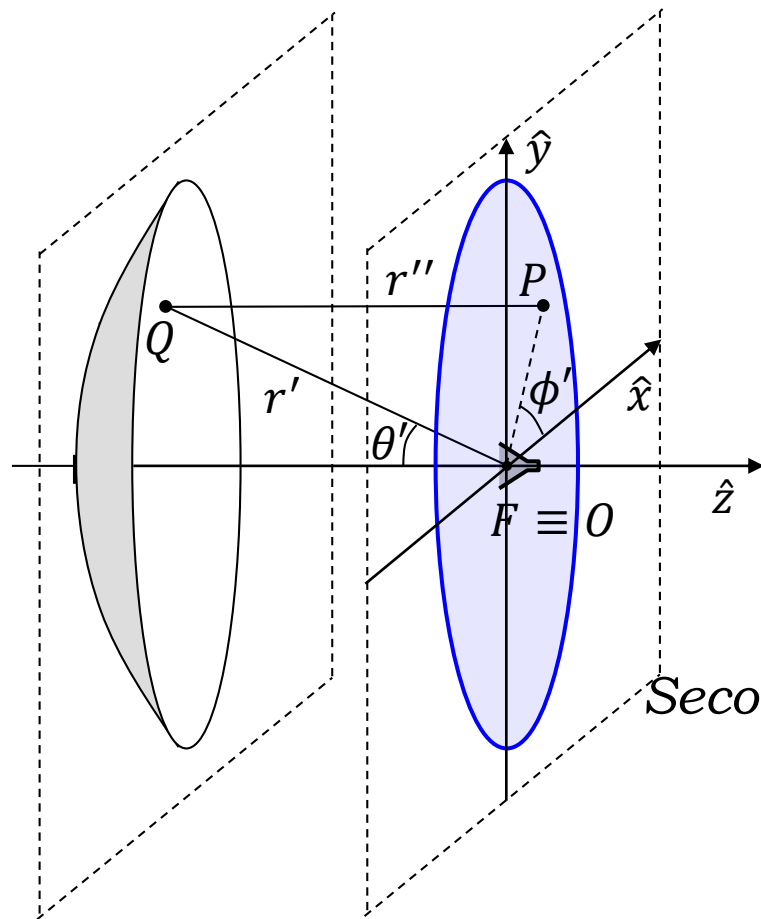
$$\frac{D}{2} = 2f \tan \left(\frac{\theta_0}{2} \right)$$

$$\theta_0 = 2 \tan^{-1} \left(\frac{D}{4f} \right)$$

Reflector f# number:

$$f_{\#} = \frac{f}{D} = \frac{1}{4 \tan \left(\frac{\theta_0}{2} \right)}$$

Analysis of the Radiation of a Parabolic Reflector



Method of the Geometrical Optics Combined to the Aperture Distribution Integral (GO+AI)

First Step: Description of the Field on the Aperture in Terms of its Phase, Amplitude, and Polarization

$$\vec{E}_r(P) = \tilde{R}(Q) \cdot \vec{E}_i(Q) S(Q) e^{-jk_0 r''}$$

Second Step: Evaluate equivalent currents

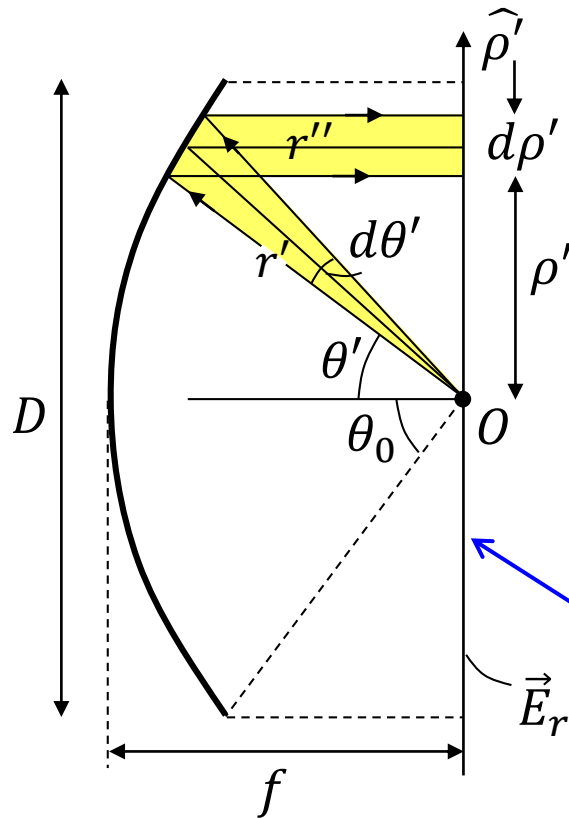
$$\vec{M}_s \approx \vec{E}_r \times \hat{n} \Big|_s$$

$$\vec{J}_s \approx \hat{n} \times \vec{H}_r \Big|_s$$

Third Step: Evaluate far field radiation

Aperture Amplitude Distribution

Find the field aperture amplitude distribution when the radiation pattern of the feed is given



The power density radiated from the feed within the solid angle $d\Omega = \sin \theta' d\theta' d\phi'$ is equal to the power density reflected by the parabola and propagating parallel to the axis z through the area $dA = \rho' d\rho' d\phi'$

$$\frac{1}{2\zeta} |\vec{E}_r(\rho', \phi')|^2 dA = U_{feed}(\theta', \phi') d\Omega$$

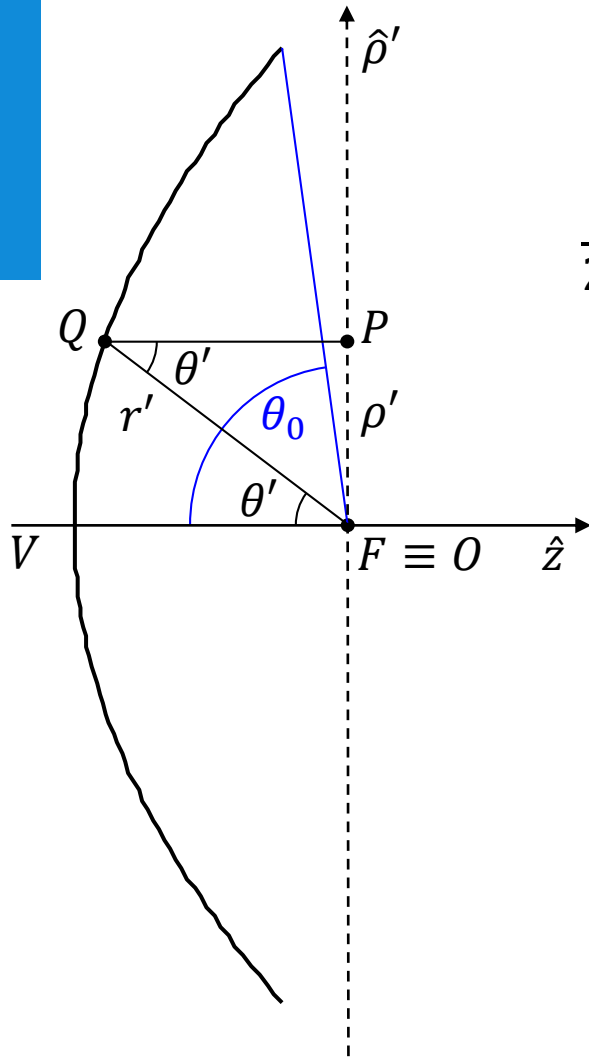
Equivalent Aperture

From Previous Lectures/EM Courses

$$U_{feed}(\theta', \phi') = \frac{P_f G_f(\theta', \phi')}{4\pi}$$

$$U_{feed}(\theta', \phi') = \frac{1}{2\zeta} |\vec{f}_{pattern}(\theta', \phi')|^2$$

Aperture Amplitude Distribution



$$\frac{1}{2\zeta} |\vec{E}_r(\rho', \phi')|^2 dA = U_{feed}(\theta', \phi') d\Omega$$

$$\frac{1}{2\zeta} |\vec{E}_r(\rho', \phi')|^2 \rho' d\rho' d\phi' = U_{feed}(\theta', \phi') \sin \theta' d\theta' d\phi'$$

$$|\vec{E}_r(\rho', \phi')|^2 = 2\zeta U_{feed}(\theta', \phi') \frac{\sin \theta'}{\rho'} \frac{d\theta'}{d\rho'}$$

$$\frac{\sin \theta'}{\rho'} = \frac{1}{r'}$$

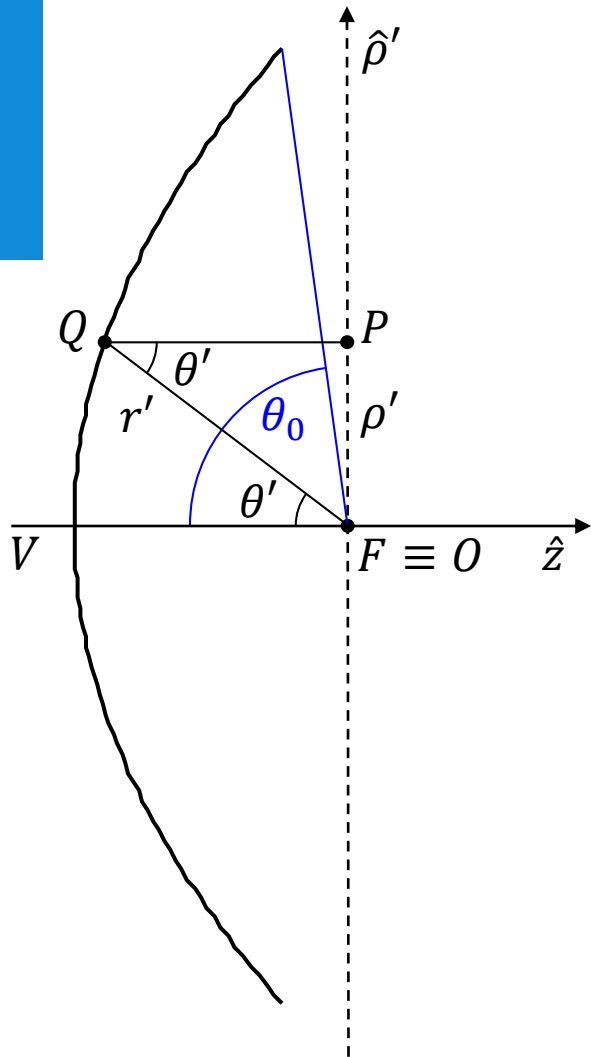
$$\rho' = 2f \tan\left(\frac{\theta'}{2}\right)$$

$$\frac{d\rho'}{d\theta'} = 2f \frac{1}{\cos^2 \frac{\theta'}{2}} \left(\frac{1}{2}\right) = \frac{f}{\cos^2 \frac{\theta'}{2}} = r'$$

$$|\vec{E}_r(\rho', \phi')|^2 = 2\zeta U_{feed}(\theta', \phi') \frac{1}{(r')^2} = |\vec{E}_i(Q)|^2$$

$$S(\rho', \phi') = 1$$

Aperture Amplitude Distribution



$$|\vec{E}_r(\rho', \phi')|^2 = 2\zeta U_{feed}(\theta', \phi') \frac{1}{(r')^2}$$

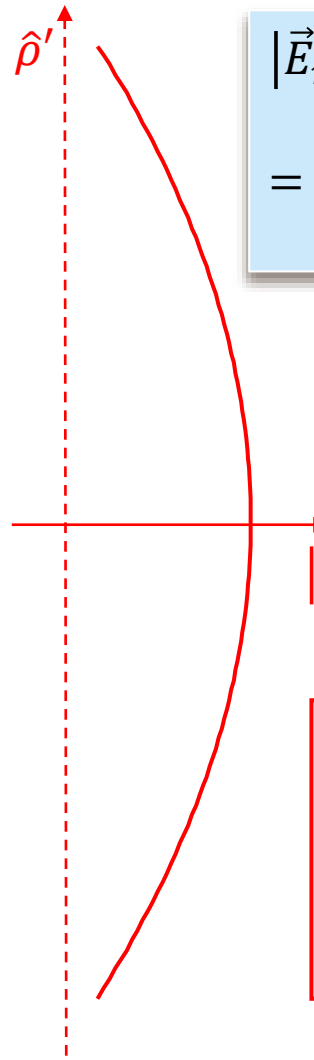
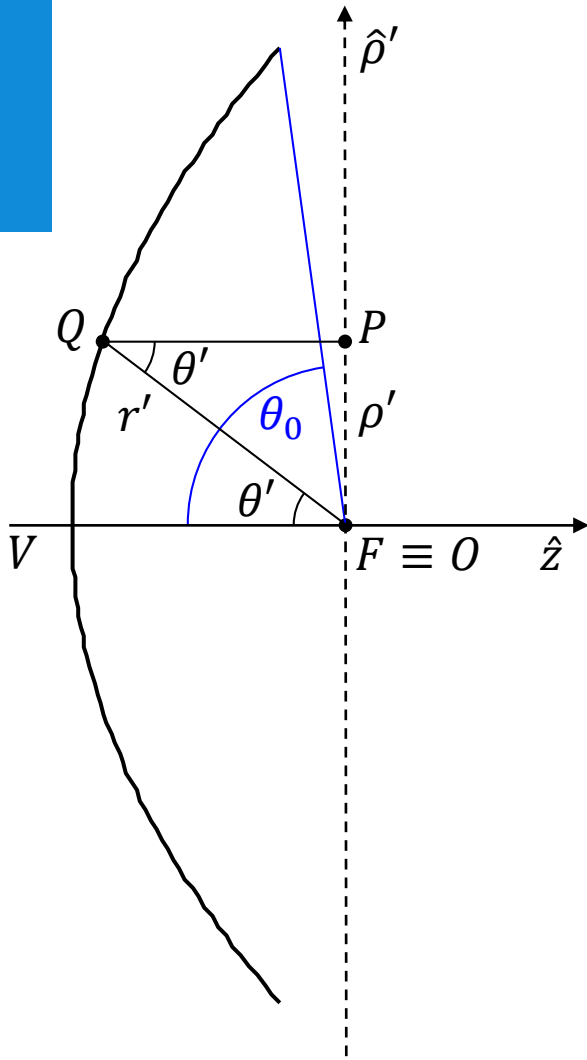
$$r' = f \left[1 + \left(\frac{\rho'}{2f} \right)^2 \right]$$

$$|\vec{E}_r(\rho', \phi')| = \sqrt{2\zeta U_{feed}(\theta', \phi')} \frac{1}{r'}$$

$$= \sqrt{2\zeta U_{feed} \left(2 \tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right)} \frac{1}{f \left[1 + \left(\frac{\rho'}{2f} \right)^2 \right]}$$

$$= \sqrt{2\zeta U_{feed} \left(2 \tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right)} \frac{4f}{4f^2 + (\rho')^2}$$

Aperture Amplitude Distribution

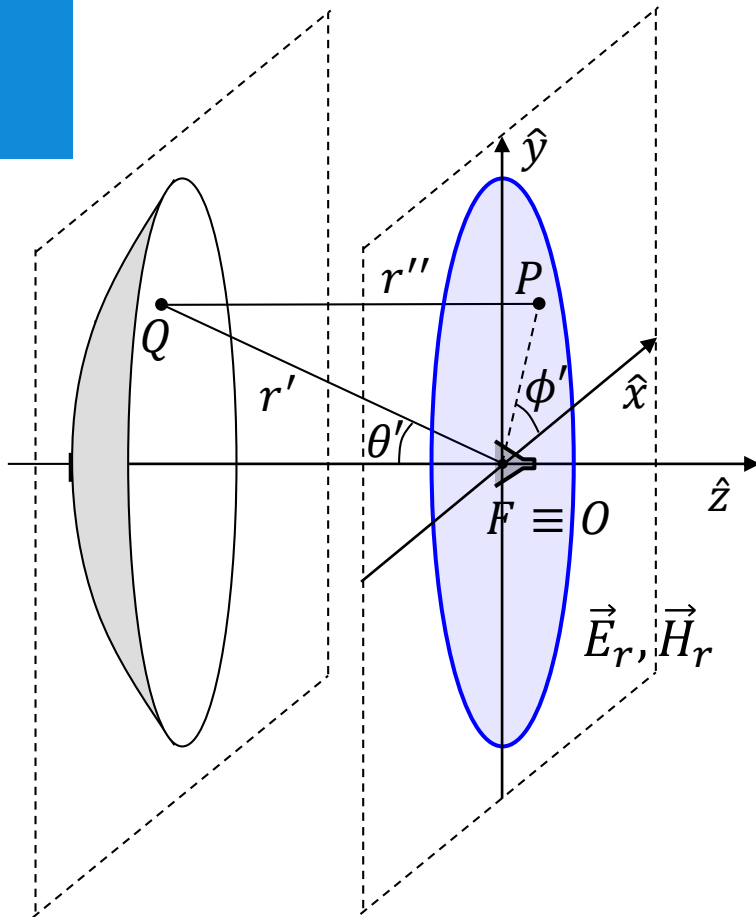


$$|\vec{E}_r(\rho', \phi')| = \sqrt{2\zeta U_{feed} \left(2\tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right) \frac{4f}{4f^2 + (\rho')^2}}$$

$$\frac{\cos^2 \left(\frac{\theta'}{2} \right)}{f}$$

If the feed radiates a uniform angular power density, the amplitude of the aperture field distribution presents a taper!

Polarization of the Reflected Rays

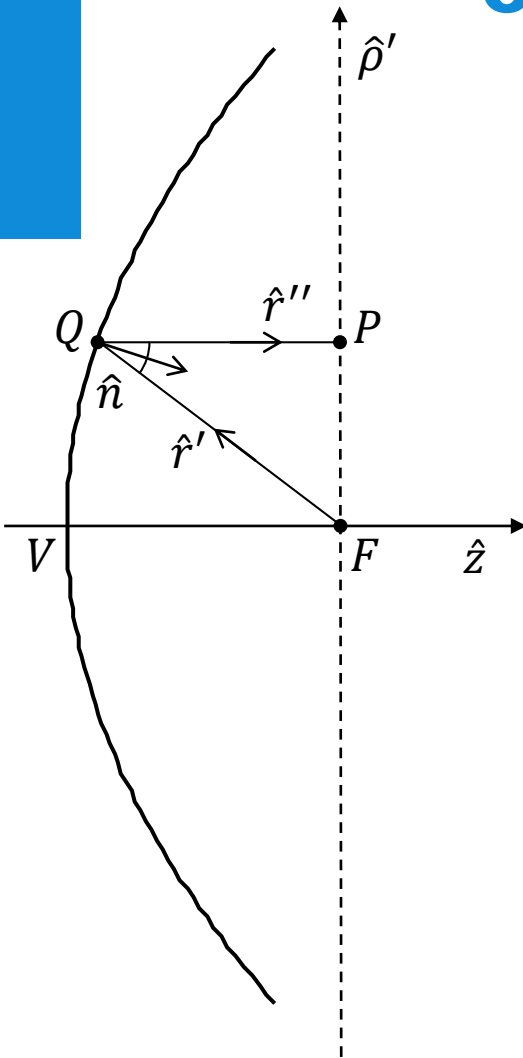


Discussed Issues:

- The fields $\vec{E}_r, \vec{H}_r$ on the aperture presents a uniform phase ($2kf$ phase due to the length of the total ray path with respect to the phase at the origin of the reference system)
- The field amplitude on the aperture can be determined by using energy flow considerations and can present in general a taper

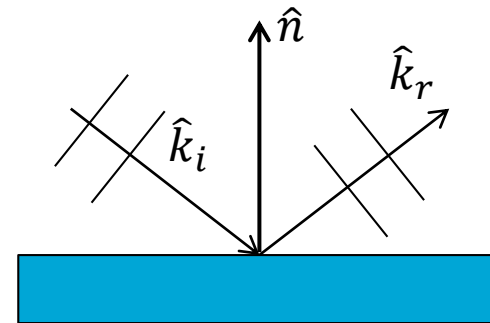
What about the polarization of the field?

Reflection Law: Propagation Constant of the Reflected Waves



$$\hat{k}_i = k\hat{r}'$$

$$\hat{k}_r = k\hat{r}''$$



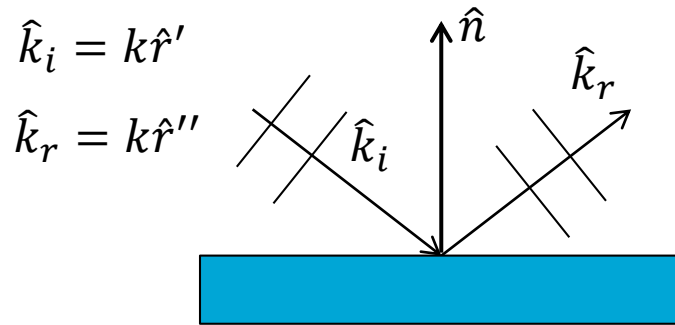
The components of $\hat{k}_r$ tangent to the planar interface are equal to the corresponding components of $\hat{k}_i$

$$1) \quad \hat{k}_i \times \hat{n} = \hat{k}_r \times \hat{n}$$

The components of $\hat{k}_r$ orthogonal to the planar interface are opposite to the corresponding components of $\hat{k}_i$

$$2) \quad \hat{k}_r \cdot \hat{n} = -\hat{k}_i \cdot \hat{n}$$

Reflection Law: Propagation Constant of the Reflected Waves



The components of $\hat{k}_r$ tangent to the planar interface are equal to the corresponding components of $\hat{k}_i$ 1) $\hat{k}_i \times \hat{n} = \hat{k}_r \times \hat{n}$

The components of $\hat{k}_r$ orthogonal to the planar interface are opposite to the corresponding components of $\hat{k}_i$ 2) $\hat{k}_r \cdot \hat{n} = -\hat{k}_i \cdot \hat{n}$

$$\hat{k}_i \times \hat{n} = \hat{k}_r \times \hat{n}$$

$$\hat{n} \times (\hat{k}_i \times \hat{n}) = \hat{n} \times (\hat{k}_r \times \hat{n})$$

$$\hat{k}_i - \hat{n}(\hat{k}_i \cdot \hat{n}) = \hat{k}_r - \hat{n}(\hat{k}_r \cdot \hat{n})$$

$$\hat{k}_i - \hat{n}(\hat{k}_i \cdot \hat{n}) = \hat{k}_r - \hat{n}(-\hat{k}_i \cdot \hat{n})$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$$

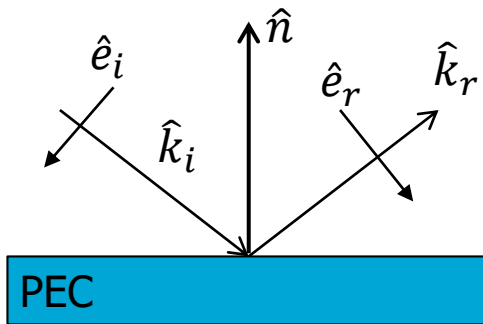
$$\hat{k}_r \cdot \hat{n} = -\hat{k}_i \cdot \hat{n}$$

$$\hat{k}_r = \hat{k}_i - 2(\hat{k}_i \cdot \hat{n})\hat{n}$$

$$\hat{r}'' = \hat{r}' - 2(\hat{r}' \cdot \hat{n})\hat{n}$$

Direction of the Reflected Ray

Reflection Law: Propagation Constant of the Reflected Waves



The components of $\vec{E}_r$ tangent to the planar interface are equal, but in opposite direction, to the corresponding components of $\vec{E}_i$

$$\vec{E}_r \times \hat{n} = -\vec{E}_i \times \hat{n}$$

The same is true for the unit vectors

$$1) \quad \hat{e}_r \times \hat{n} = -\hat{e}_i \times \hat{n}$$

The components of $\hat{e}_r$ orthogonal to the planar interface are equal to the corresponding components of $\hat{e}_i$

$$2) \quad \hat{e}_r \cdot \hat{n} = \hat{e}_i \cdot \hat{n}$$

$$\hat{e}_r \times \hat{n} = -\hat{e}_i \times \hat{n}$$

$$\hat{n} \times (\hat{e}_r \times \hat{n}) = -\hat{n} \times (\hat{e}_i \times \hat{n})$$

$$\hat{e}_r - \hat{n}(\hat{e}_r \cdot \hat{n}) = -\hat{e}_i + \hat{n}(\hat{e}_i \cdot \hat{n})$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$$

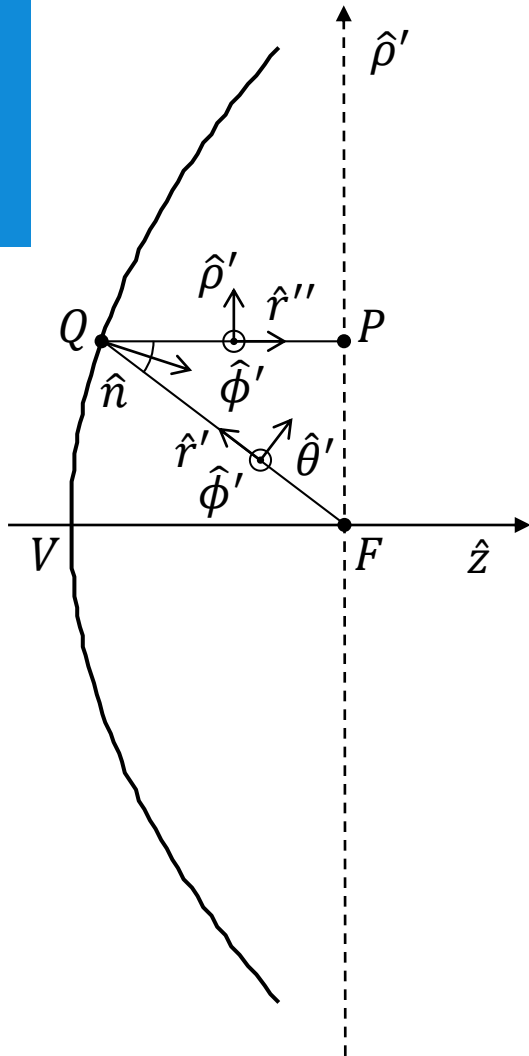
$$\hat{e}_r - \hat{n}(\hat{e}_i \cdot \hat{n}) = -\hat{e}_i + \hat{n}(\hat{e}_i \cdot \hat{n})$$

$$\hat{e}_r \cdot \hat{n} = \hat{e}_i \cdot \hat{n}$$

$$\hat{e}_r = -\hat{e}_i + 2(\hat{e}_i \cdot \hat{n})\hat{n}$$

Polarization of the Reflected Ray

Unit Normal Vector to a Parabolic Reflector



$$\hat{n} = -\cos\left(\frac{\theta'}{2}\right)\hat{r}' + \sin\left(\frac{\theta'}{2}\right)\hat{\theta}'$$

$$z = -f + \frac{x^2 + y^2}{4f}$$

$$\hat{n} = \frac{-\frac{x}{2f}\hat{x} - \frac{y}{2f}\hat{y} + \hat{z}}{\sqrt{\left(\frac{x}{2f}\right)^2 + \left(\frac{y}{2f}\right)^2 + 1}}$$

Geometrical Optics Ray Reference Systems

GO Incident Ray Reference System

$$\hat{r}' = \frac{Q - F}{|Q - F|}$$

$$\hat{\phi}' = \frac{\hat{n} \times \hat{r}'}{|\hat{n} \times \hat{r}'|}$$

$$\hat{\theta}' = \hat{r}' \times \hat{\phi}'$$

$$\hat{e}_i = e_{i\phi'} \hat{\phi}' + e_{i\theta'} \hat{\theta}'$$

GO Reflected Ray Reference System

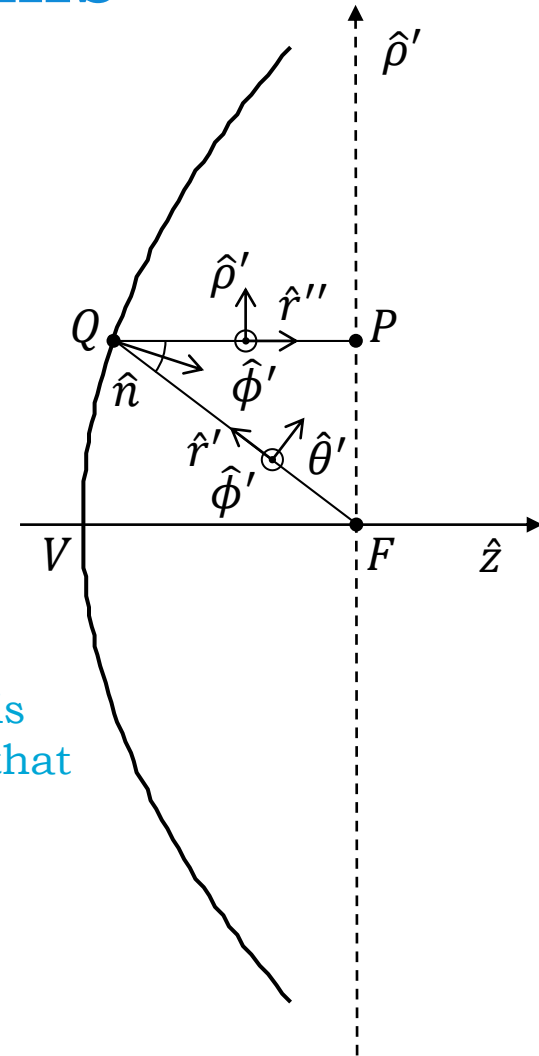
$$\hat{r}'' = \hat{r}' - 2(\hat{r}' \cdot \hat{n})\hat{n}$$

$$\hat{\phi}' = \frac{\hat{n} \times \hat{r}'}{|\hat{n} \times \hat{r}'|} \equiv \frac{\hat{n} \times \hat{r}''}{|\hat{n} \times \hat{r}''|}$$

$$\hat{\rho}' = \hat{\phi}' \times \hat{r}'' \equiv \hat{\phi}' \times \hat{z}$$

From reflection law, it is possible to demonstrate that (see Appendix)

$$\begin{aligned} \hat{e}_r &= e_{r\phi'} \hat{\phi}' + e_{r\rho'} \hat{\rho}' \\ &= -e_{i\phi'} \hat{\phi}' - e_{i\theta'} \hat{\theta}' \end{aligned}$$



Geometrical Optics Reflected Field Compact Dyadic Notation

Reflection Dyads for the Chosen Ray Reference Systems

$$\tilde{R} = -\hat{\rho}'\hat{\theta}' - \hat{\phi}'\hat{\phi}'$$

$$\begin{aligned}\hat{e}_r &= e_{r\phi'}\hat{\phi}' + e_{r\rho'}\hat{\rho}' \\ &= -e_{i\phi'}\hat{\phi}' - e_{i\theta'}\hat{\rho}'\end{aligned}$$

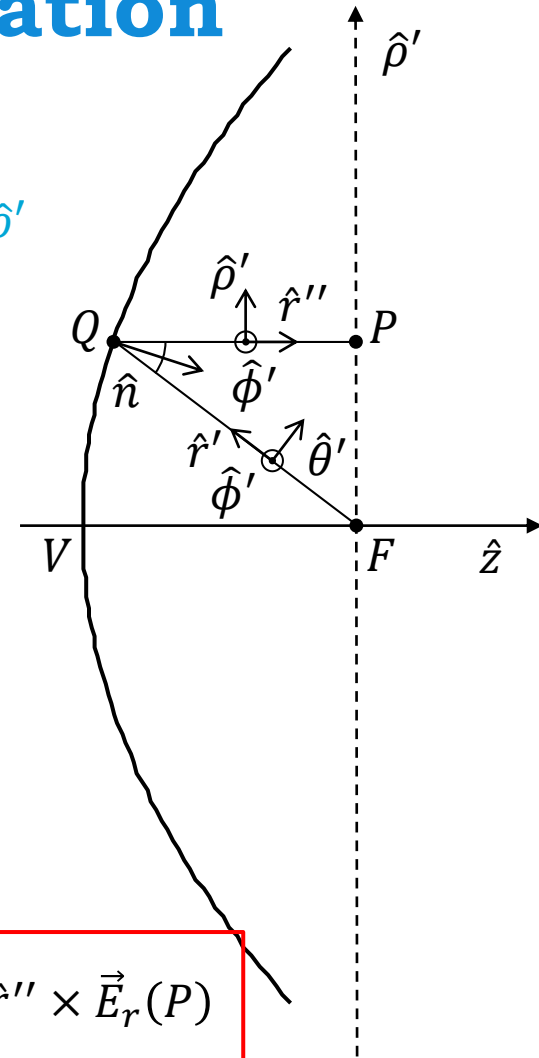
GO Reflected Ray Field

$$\text{In } P \quad \vec{E}_r(P) = \tilde{R}(Q) \cdot \vec{E}_i(Q) S(Q) e^{-jkr''}$$

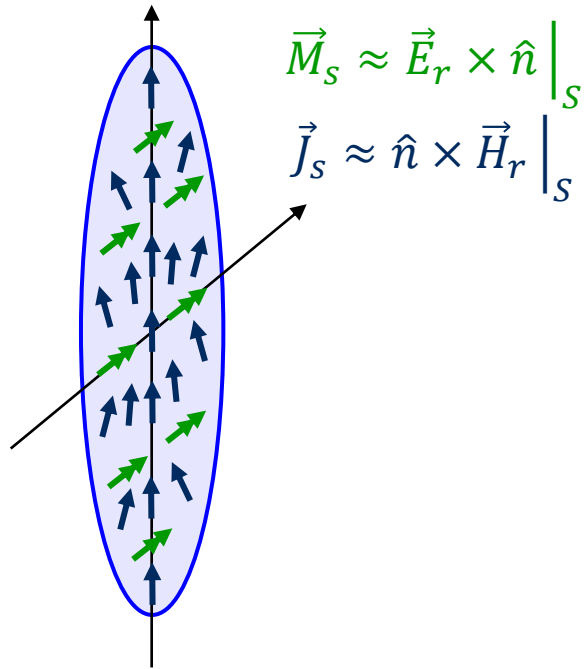
$$\text{In } P \quad \begin{bmatrix} E_{r,\rho'}(P) \\ E_{r,\phi'}(P) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} E_{i,\theta'}(Q) \\ E_{i,\phi'}(Q) \end{bmatrix} e^{-jkr''}$$

Expression valid for an incident spherical wavefront whose center is located at the parabola focus.

$$\vec{H}_r(P) = \frac{1}{\zeta} \hat{r}'' \times \vec{E}_r(P)$$



Aperture Equivalent Currents



$$\vec{E}_r(P) = \tilde{R}(Q) \cdot \vec{E}_{feed}(Q) e^{-jk_0 r''}$$

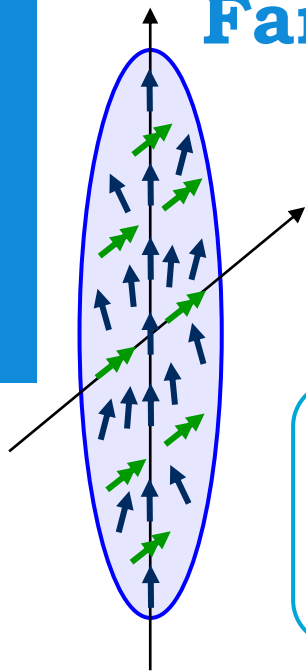
$$\vec{E}_{feed} = \left(E_{feed}^{\theta} \hat{\theta} + E_{feed}^{\phi} \hat{\phi} \right) \frac{e^{-jk_0 r'}}{r'}$$

$$\text{Phase: } r' + r'' = 2f$$

$$\text{Amplitude: } \frac{1}{r'} = \frac{1}{f \left[1 + \left(\frac{\rho'}{2f} \right)^2 \right]} = \frac{\cos^2 \left(\frac{\theta'}{2} \right)}{f}$$

$$\vec{M}_s \approx \vec{E}_r \times \hat{z} \Big|_s = \left[\left(-E_{feed}^{\theta} \hat{\rho} - E_{feed}^{\phi} \hat{\phi} \right) e^{-jk_0 2f} \frac{\cos^2 \left(\frac{\theta'}{2} \right)}{f} \right] \times \hat{z}$$

Far Field Estimation: Radiation of the Equivalent Currents



$$\begin{aligned}\vec{M} &= \vec{E}_r \times \hat{n} \Big|_S \\ \vec{J} &= \hat{n} \times \vec{H}_r \Big|_S\end{aligned}$$

From Previous Lectures/EM Courses:
Far Field Estimation – Spectral Approach

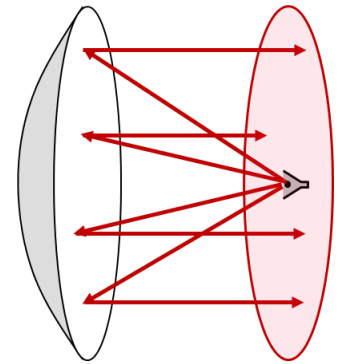
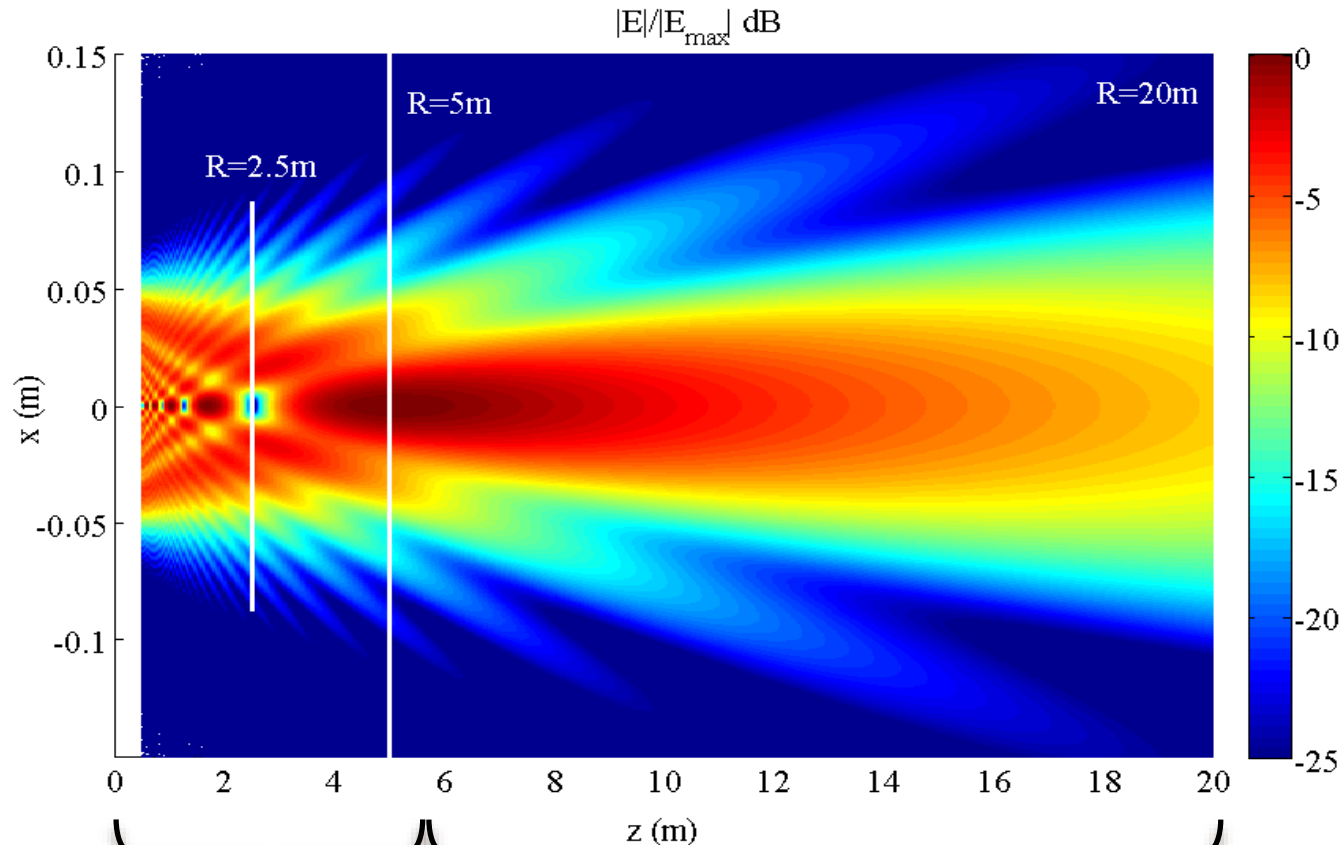
$$\vec{f}^{far}(\vec{r}) \approx jk_{zs} \tilde{\mathbf{G}}^{fc}(k_{xs}, k_{ys}, z, z') \vec{\mathcal{C}}(k_{xs}, k_{ys}) \frac{e^{-jkr}}{2\pi r}$$

From Previous Lectures/EM Courses:
Far Field Estimation – Spatial Approach

$$\begin{aligned}\vec{E}^{far}(\vec{r}) &\approx \left[-\omega\mu(\tilde{I} - \hat{r}\hat{r}) \iint_S \vec{J}(\vec{r}') e^{jk\hat{r}\cdot\vec{r}'} dS + k\hat{r} \times \tilde{I} \iint_S \vec{M}(\vec{r}') e^{jk\hat{r}\cdot\vec{r}'} dS \right] \frac{je^{-jkr}}{4\pi r} \\ \vec{H}^{far}(\vec{r}) &\approx \left[-\omega\varepsilon(\tilde{I} - \hat{r}\hat{r}) \iint_S \vec{M}(\vec{r}') e^{jk\hat{r}\cdot\vec{r}'} dS - k\hat{r} \times \tilde{I} \iint_S \vec{J}(\vec{r}') e^{jk\hat{r}\cdot\vec{r}'} dS \right] \frac{je^{-jkr}}{4\pi r}\end{aligned}$$

Radiation from large apertures

the field radiated at
600GHz by a 10cm dish

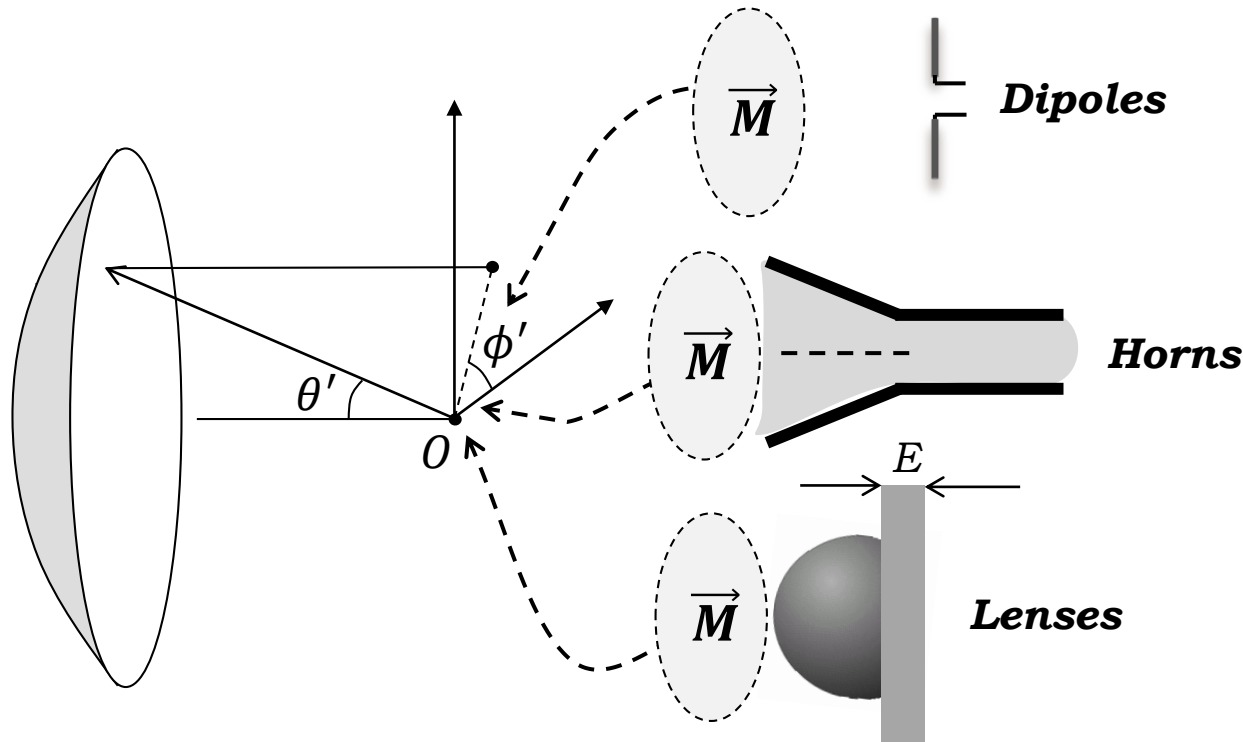


The GO ray
field picture is
only valid very
close to the
reflector

Fraunhofer
region at

$$R \leq \frac{2D^2}{\lambda} = 40m$$

Typical Feeds for Reflectors

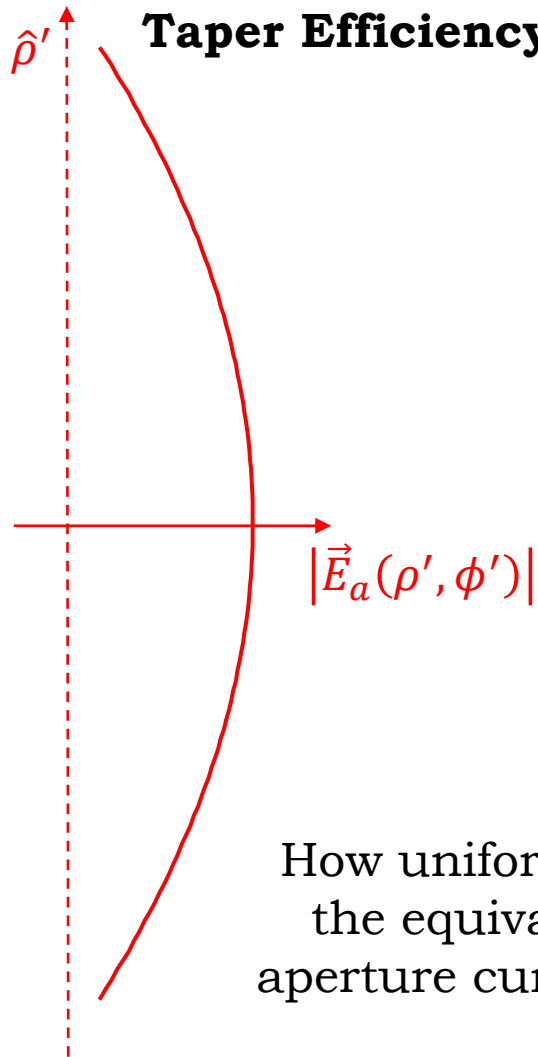


$$\vec{E}(k_x, k_y) = -2j \frac{k e^{-jkr'}}{4\pi r'} \vec{M} \times \hat{r}' \quad \text{Feed Pattern Spectral Representation Model}$$

$$\vec{E}(Q) = \vec{f}_{pattern}(\theta', \phi') \frac{e^{-jkr'}}{r'} \quad \text{Feed Pattern Spatial Representation Model}$$

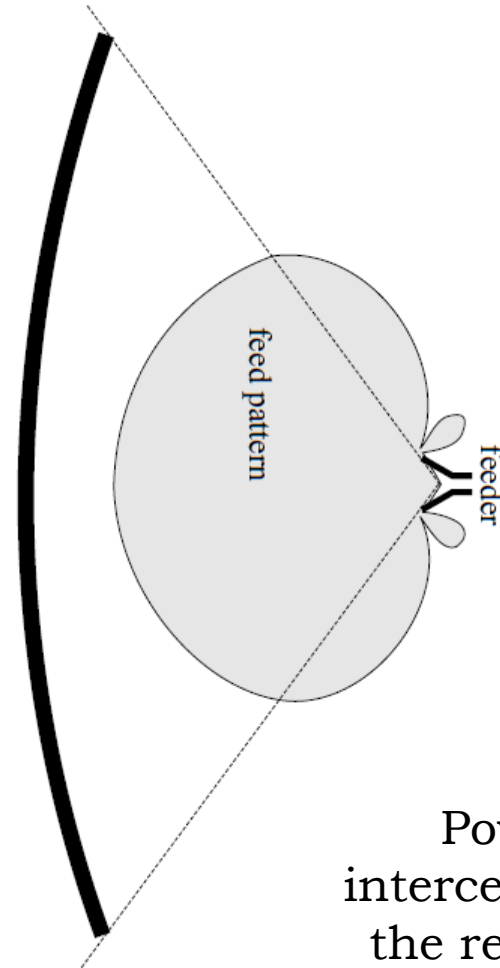
Reflector Antenna Efficiencies

Taper Efficiency



How uniform are
the equivalent
aperture currents?

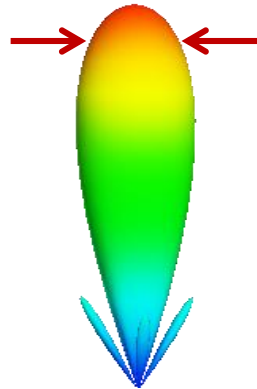
Spillover Efficiency



Power
intercepted by
the reflector

Taper Efficiency

Large reflectors achieve directive beams that enable angular discrimination



$\Delta\theta_{HPBW}$

The HPBW defines the angular resolution of the system

Uniform current distribution



$$D_{ir}^{max} = \frac{4\pi\pi\left(\frac{D}{2}\right)^2}{\lambda_0^2} = \frac{\pi^2 D^2}{\lambda_0^2}$$

$$\Delta\theta_{HPBW}^{min} = \frac{\lambda_0}{D}$$

For a feed on axis

The taper efficiency gives an estimation of how good is this angular discrimination

$$\eta_t = \frac{D_{ir}(0,0)}{D_{ir}^{max}}$$

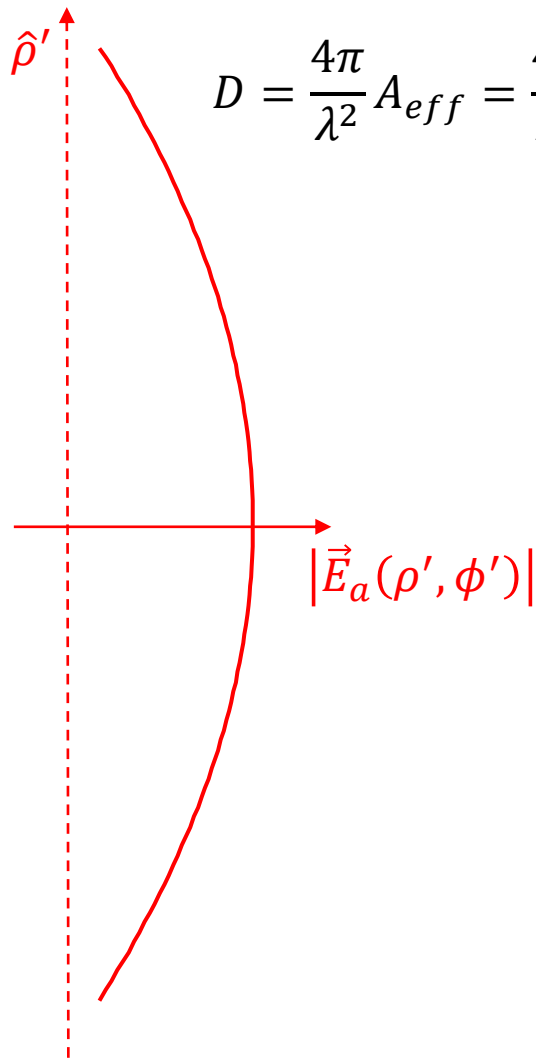
$$D_{ir}(0,0) \approx \frac{\pi^2}{(\Delta\theta_{HPBW})^2}$$



$$\eta_t \approx \left(\frac{\Delta\theta_{HPBW}^{min}}{\Delta\theta_{HPBW}} \right)^2$$

Taper Efficiency: Aperture field

For a feed on axis



$$D = \frac{4\pi}{\lambda^2} A_{eff} = \frac{4\pi}{\lambda^2} \eta_t A \quad A_{eff} = \frac{\left| \iint_A E_a^{co}(\rho', \phi') dA \right|^2}{\iint_A |\vec{E}_a(\rho', \phi')|^2 dA}$$

$$\eta_t = \frac{A_{eff}}{A} = \frac{1}{A} \frac{\left| \iint_A E_a^{co}(\rho', \phi') dA \right|^2}{\iint_A |\vec{E}_a(\rho', \phi')|^2 dA}$$

Function of the Aperture Field

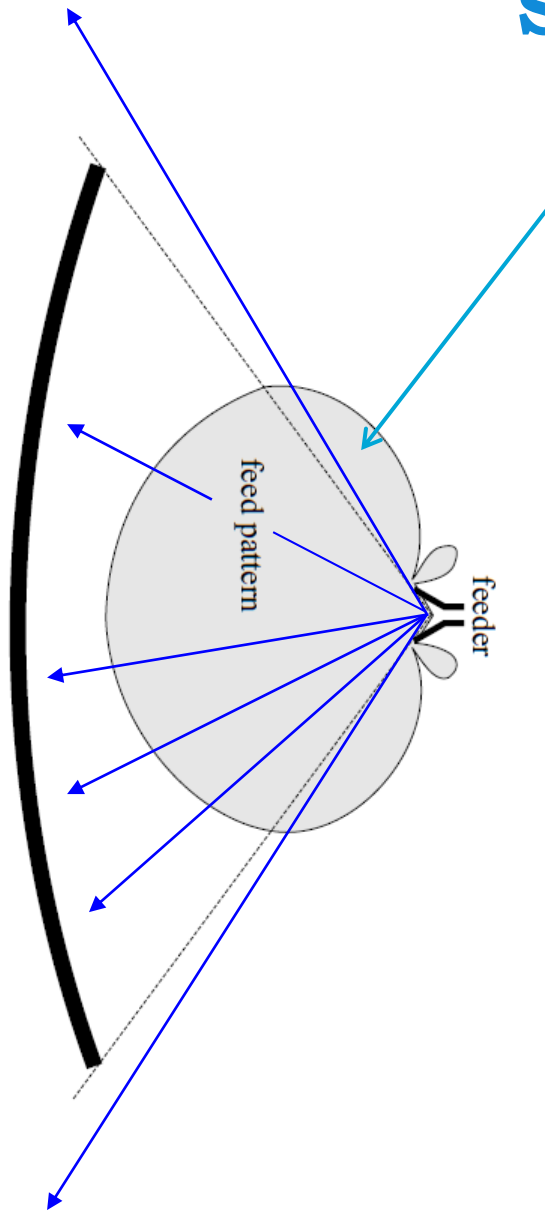
Function of the Feed Gain

$$\eta_t = \frac{16}{\pi} \left(\frac{f}{D} \right)^2 \frac{\left| \int_0^{2\pi} \int_0^{\theta_0} \sqrt{G_{co}(\theta', \phi')} \tan\left(\frac{\theta'}{2}\right) d\theta' d\phi' \right|^2}{\int_0^{2\pi} \int_0^{\theta_0} G(\theta', \phi') \sin \theta' d\theta' d\phi'}$$

For circularly symmetric gain feeds

$$\eta_t = 32 \left(\frac{f}{D} \right)^2 \frac{\left| \int_0^{\theta_0} \sqrt{G_{co}(\theta')} \tan\left(\frac{\theta'}{2}\right) d\theta' \right|^2}{\int_0^{\theta_0} G(\theta') \sin \theta' d\theta'}$$

Spillover Efficiency



Measure of the feed power which is not captured by the reflector

$$\eta_s = \frac{\int_0^{2\pi} \int_0^{\theta_0} U_{feed}(\theta', \phi') \sin \theta' d\theta' d\phi'}{\int_0^{2\pi} \int_0^{\pi} U_{feed}(\theta', \phi') \sin \theta' d\theta' d\phi'}$$

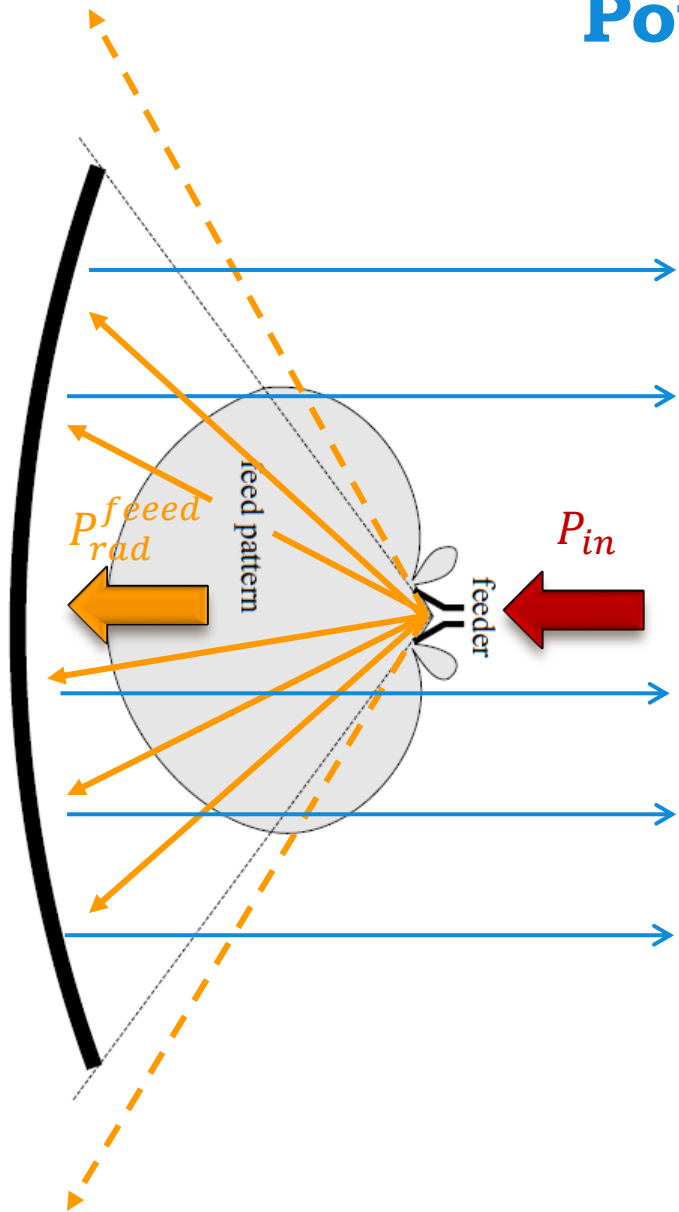
Function of the Feed Gain

$$\eta_s = \frac{\int_0^{2\pi} \int_0^{\theta_0} G_f(\theta', \phi') \sin \theta' d\theta' d\phi'}{\int_0^{2\pi} \int_0^{\pi} G_f(\theta', \phi') \sin \theta' d\theta' d\phi'}$$

For circularly symmetric gain feeds

$$\eta_s = \frac{\int_0^{\theta_0} G_f(\theta') \sin \theta' d\theta'}{\int_0^{\pi} G_f(\theta') \sin \theta' d\theta'}$$

Power Budget

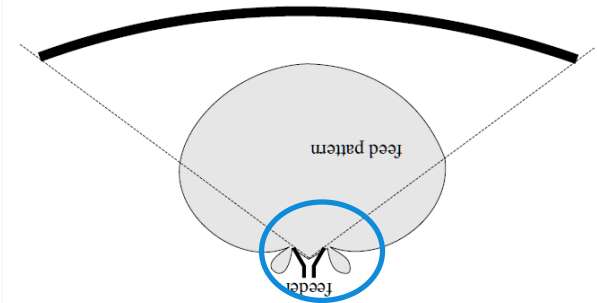
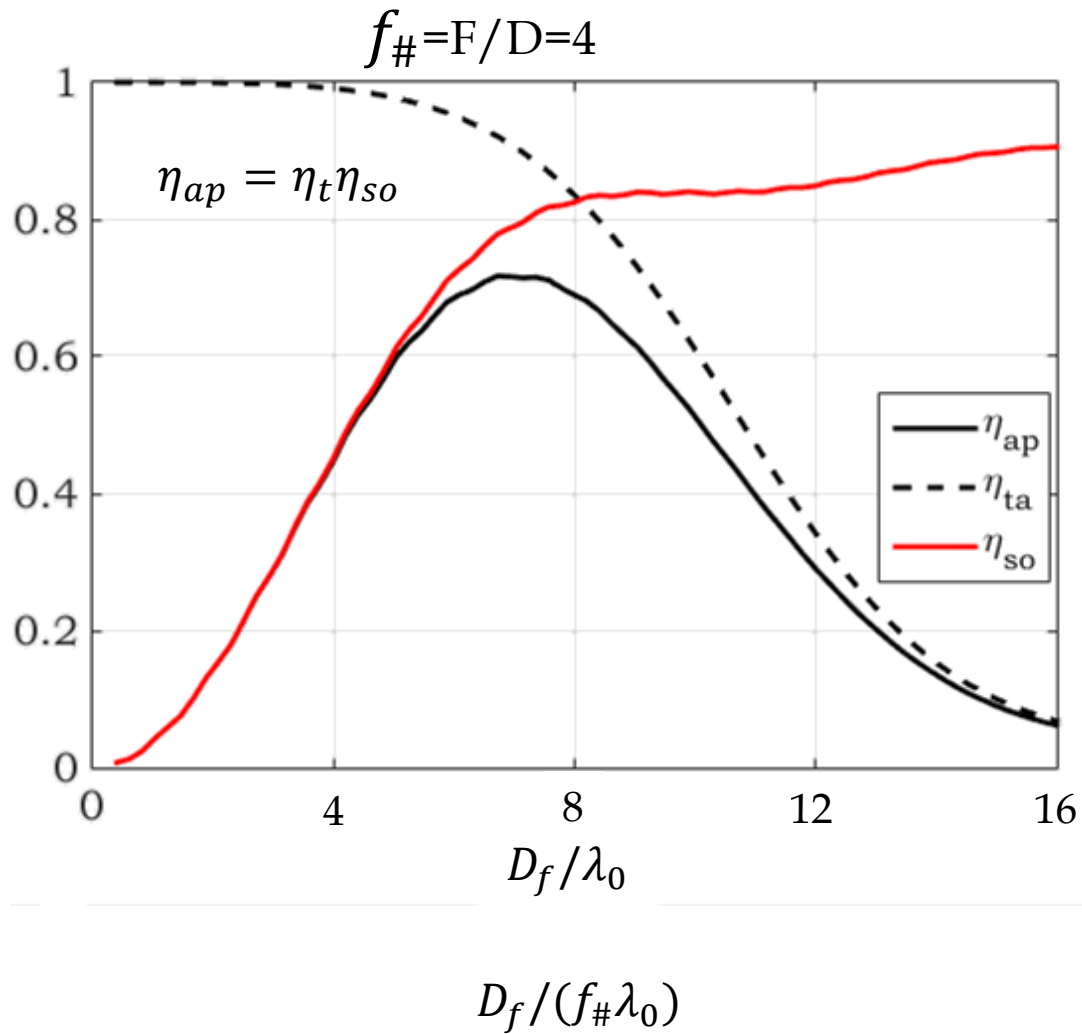


$$P_{rad}^{ref} = \eta_{so} P_{rad}^{feed} = \eta_{so} \eta_{feed} P_{in}$$

$$\eta_{so} = \frac{P_{rad}^{ref}}{P_{rad}^{feed}} \quad \eta_{feed} = \frac{P_{rad}^{feed}}{P_{in}}$$

$$G = \eta_{so} \eta_{feed} D$$

Reflector Aperture Efficiency: Example



Feed

$$\vec{J}_{eq} = \text{circ} \left(\rho, \frac{D_f}{2} \right) \hat{x}$$

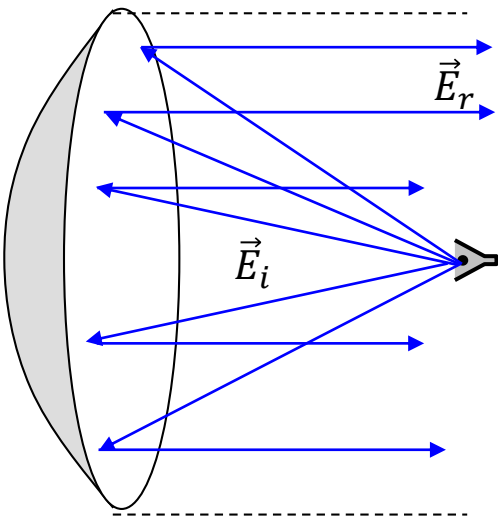
D_f

Feed with
uniform current
distribution

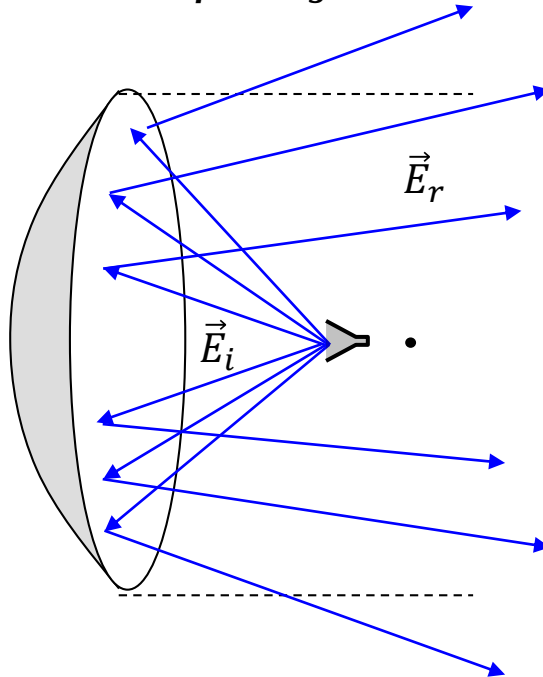
Effects of the Feed Defocusing

Let us try to follow some of the GO reflected rays

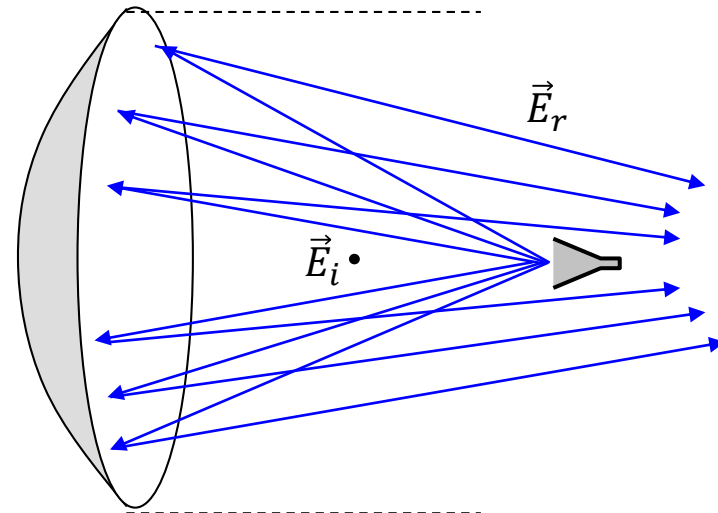
*Feed in focus of parabola,
Focus at infinity*



*Advanced focus,
defocusing beam*



*Retreat of the focus,
near field focus*

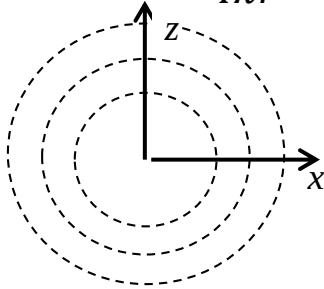


More realistically, you will try to put antenna in focus, but you will not manage, there is going to be a phase error that causes reduced directivity with respect to maximum defined by physical area

Feed Phase Center Location

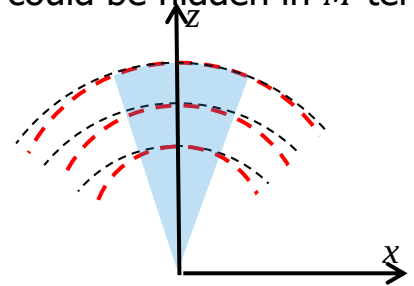
An explicit phase center is in origin of reference system

$$(1) \vec{E}(k_x, k_y) = -2j \frac{k e^{-jkr}}{4\pi r} \vec{M} \times \hat{r}$$



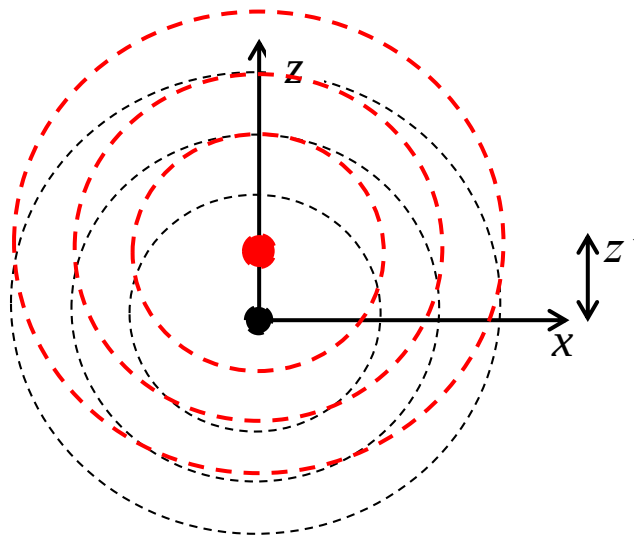
However different phase center could be hidden in $\vec{M}$ terms

One could realize that the phase can be approximated as spherical only over a small angular region



Actually this is what happens always unless you are incredibly lucky

For antenna centered in origin O
One can use eq. (1)



For an antenna located in z' ,
if one wishes to use (1)
he should explicitly include the
phase term

$$e^{\pm jk_z z'};$$

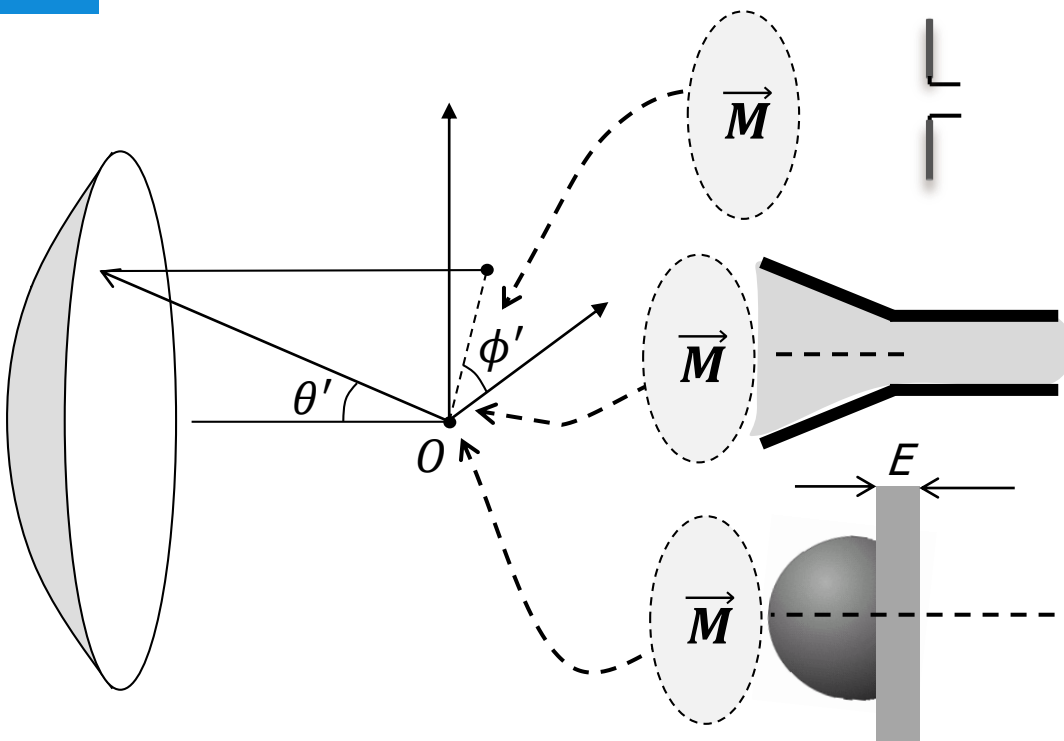
+ for $z > z'$ and - for $z < z'$

Feed Phase Center Location

An explicit phase center is in origin of reference system

$$\vec{E}(k_x, k_y) = -2j \frac{ke^{-jkr}}{4\pi r} \vec{M} \times \hat{r}$$

One should try to put the antenna feed phase center in the geometrical focus, it's trial and error unless one knows exact phase center



Dipoles

No problem: they are small in terms of wavelength

Horns

Big Problem, phase center, is somewhere in the throat
Is function of the frequency

Lenses

Phase center depends on extension length
and the feed.

If they are broad band,
phase center moves a lot!

Other Efficiencies – Quick Summary

We discussed:

- **Taper Efficiency** η_t
- **Spillover Efficiency** η_s
- **Aperture Efficiency** η_{ap}

Moreover:

- **Phase uniformity of the field over the aperture plane (generally known as phase efficiency η_p)**
- **Polarization uniformity of the field over the aperture plane (generally known as polarization efficiency η_x)**
- **Blockage Efficiency η_b**
- **Random Error Efficiency η_r (Surface roughness, Distortions of the Reflector)**
- **Other ...**

In general the Aperture Efficiency can be written as: $\eta_{ap} = \eta_t \eta_s \eta_p \eta_x \eta_b \eta_r$

Truly Important (also for the assignments): Equivalent Aperture Field Distribution

$$\vec{E}_a(\rho', \phi') \approx \vec{E}_r(\rho', \phi') = \sqrt{2\zeta U_{feed} \left(2 \tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right)} \frac{4f}{4f^2 + (\rho')^2} e^{-j2kf} \hat{e}_r$$

$$\hat{e}_r = \bar{R} \cdot \hat{e}_i$$

$$\bar{R} = -\hat{\rho}'\hat{\theta}' - \hat{\phi}'\hat{\phi}'$$

$$\hat{e}_i = e_{i\phi'}\hat{\phi}' + e_{i\theta'}\hat{\theta}'$$

$$\vec{H}_a(\rho', \phi') \approx \vec{H}_r(\rho', \phi') = \frac{1}{\zeta} \hat{r}'' \times \vec{E}_r(\rho', \phi') = \frac{1}{\zeta} \hat{z} \times \vec{E}_r(\rho', \phi')$$

Truly Important (also for the assignments): Efficiencies & Gain

$$\eta_t = \frac{A_{eff}}{A} = \frac{1}{A} \frac{\left| \iint_A E_a^{co}(\rho', \phi') dA \right|^2}{\iint_A |\vec{E}_a(\rho', \phi')|^2 dA}$$

$$\eta_s = \frac{\int_0^{2\pi} \int_0^{\theta_0} U_{feed}(\theta', \phi') \sin \theta' d\theta' d\phi'}{\int_0^{2\pi} \int_0^{\pi} U_{feed}(\theta', \phi') \sin \theta' d\theta' d\phi'}$$

$$\eta_{ap} = \eta_s \eta_t$$

$$G = \frac{4\pi}{\lambda^2} \eta_{ap} A = \frac{4\pi}{\lambda^2} \eta_s \eta_t A$$

$$D_M = \frac{4\pi}{\lambda^2} A$$

Thanks for your Attention!



Arecibo Observatory, Puerto Rico

Appendix: Polarization of the Reflected Field

Polarization of the incident field: $\hat{e}_i = e_{i\phi'}\hat{\phi}' + e_{i\theta'}\hat{\theta}'$

Polarization of the reflected field: $\hat{e}_r = e_{r\phi'}\hat{\phi}' + e_{r\rho'}\hat{\rho}'$

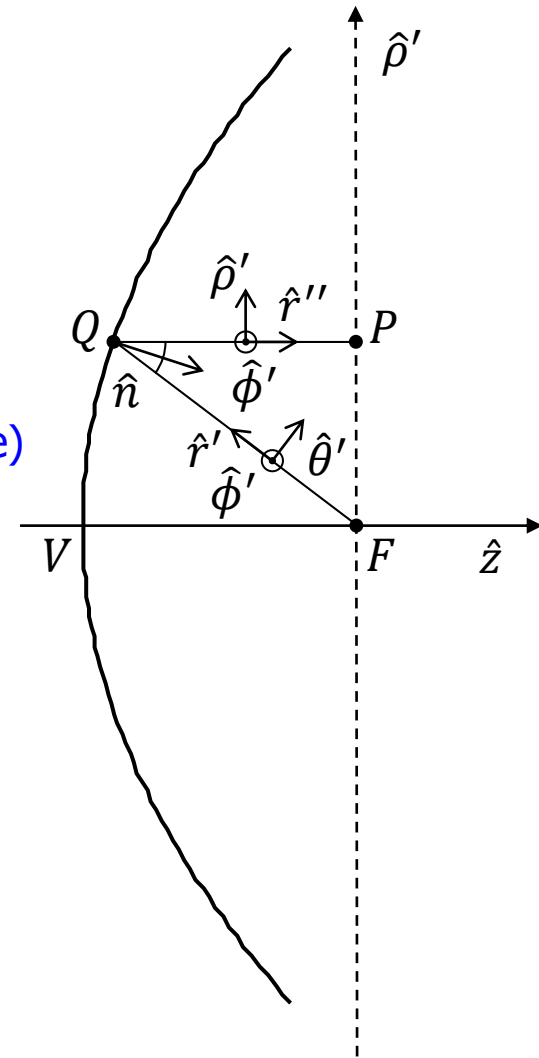
Perpendicular Polarization: $\hat{e}_i = e_{i\phi'}\hat{\phi}', \hat{e}_r = e_{r\phi'}\hat{\phi}'$

$\hat{e}_r = -\hat{e}_i + 2(\hat{e}_i \cdot \hat{n})\hat{n}$ Polarization of the Reflected Ray (PEC Surface)

$$\hat{e}_r = -\hat{e}_i + 2(\hat{e}_i \cdot \hat{n})\hat{n} = -e_{i\phi'}\hat{\phi}' + 2[(e_{i\phi'}\hat{\phi}') \cdot \hat{n}]\hat{n}$$

$$\hat{e}_r = -e_{i\phi'}\hat{\phi}' \quad \text{Since } \hat{\phi}' \cdot \hat{n} = 0$$

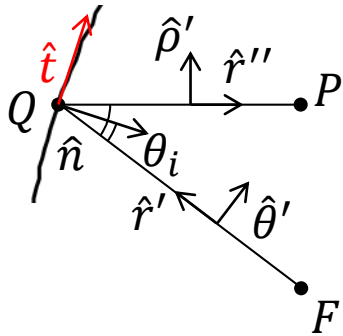
$$e_{r\phi'} = -e_{i\phi'}$$



Appendix: Polarization of the Reflected Field

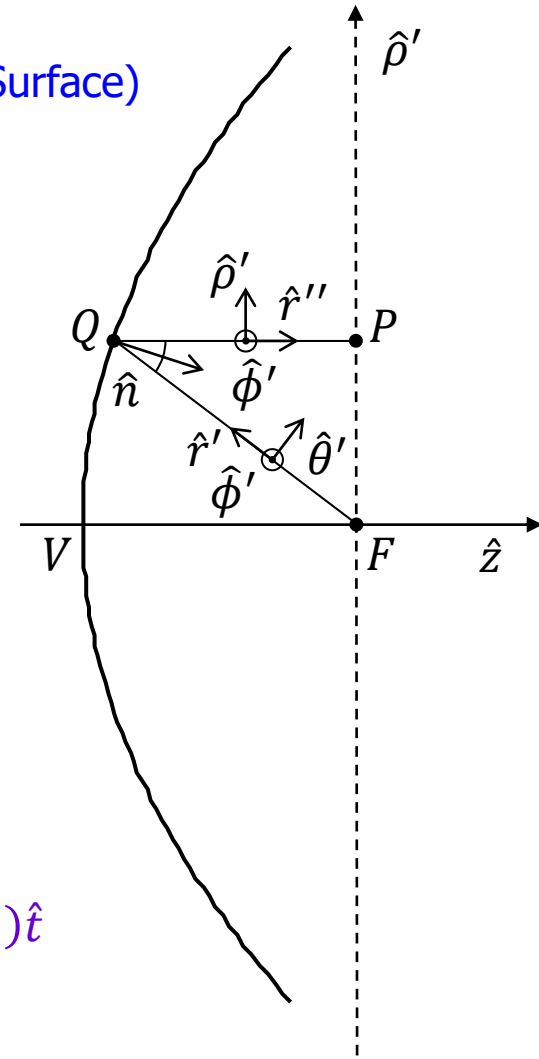
Parallel Polarization: $\hat{e}_i = e_{i\theta'}\hat{\theta}', \hat{e}_r = e_{r\rho'}\hat{\rho}'$

$\hat{e}_r = -\hat{e}_i + 2(\hat{e}_i \cdot \hat{n})\hat{n}$ Polarization of the Reflected Ray (PEC Surface)



$\hat{t}$ such that $\hat{t} \cdot \hat{n} = 0$ $\hat{n} \times \hat{t} = \hat{\phi}'$

θ_i such that $\cos(\theta_i) = -\hat{r}' \cdot \hat{n}$



$\hat{\rho}'$ in terms of $\hat{n}$ and $\hat{t}$:

$$\hat{\rho}' = (\hat{\rho}' \cdot \hat{n})\hat{n} + (\hat{\rho}' \cdot \hat{t})\hat{t}$$

$$\hat{\rho}' = \cos\left(\theta_i + \frac{\pi}{2}\right)\hat{n} + \cos(\theta_i)\hat{t}$$

$$\hat{\rho}' = -\sin(\theta_i)\hat{n} + \cos(\theta_i)\hat{t}$$

$\hat{\theta}'$ in terms of $\hat{n}$ and $\hat{t}$:

$$\hat{\theta}' = (\hat{\theta}' \cdot \hat{n})\hat{n} + (\hat{\theta}' \cdot \hat{t})\hat{t}$$

$$\hat{\theta}' = \cos\left(\frac{\pi}{2} - \theta_i\right)\hat{n} + \cos(\theta_i)\hat{t}$$

$$\hat{\theta}' = \sin(\theta_i)\hat{n} + \cos(\theta_i)\hat{t}$$

Appendix: Polarization of the Reflected Field

Parallel Polarization: $\hat{e}_i = e_{i\theta'}\hat{\theta}', \hat{e}_r = e_{r\rho'}\hat{\rho}'$

$\hat{e}_r = -\hat{e}_i + 2(\hat{e}_i \cdot \hat{n})\hat{n}$ Polarization of the Reflected Ray (PEC Surface)

$$\hat{e}_r = -\hat{e}_i + 2(\hat{e}_i \cdot \hat{n})\hat{n} = -e_{i\theta'}\hat{\theta}' + 2[(e_{i\theta'}\hat{\theta}') \cdot \hat{n}]\hat{n}$$

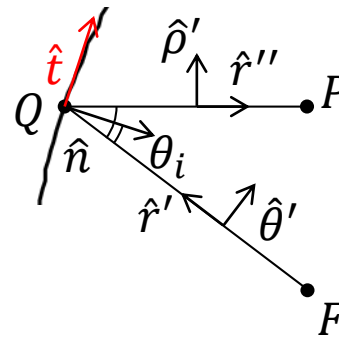
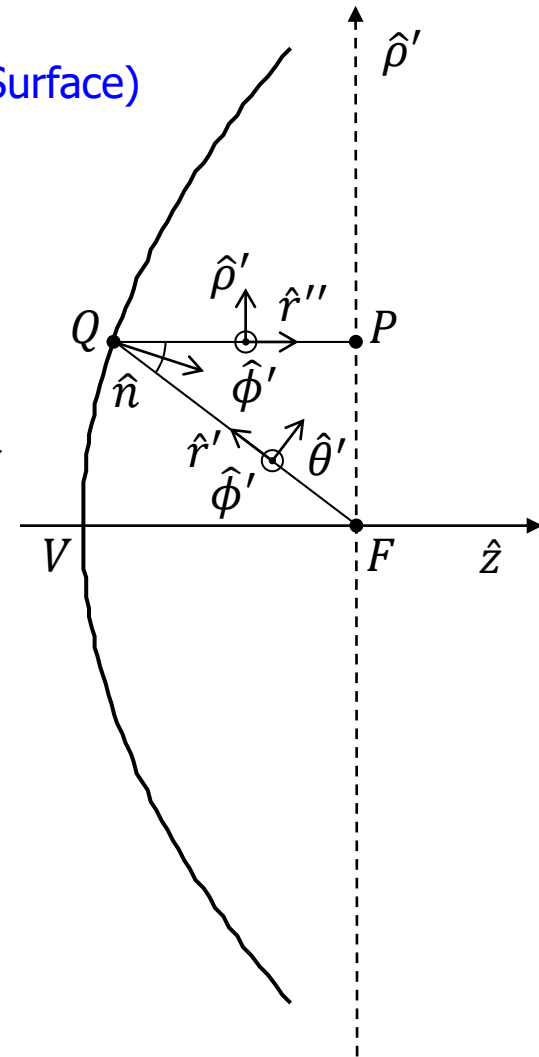
$$\hat{e}_r = -e_{i\theta'}[\sin(\theta_i)\hat{n} + \cos(\theta_i)\hat{t}] + 2e_{i\theta'}\{[\sin(\theta_i)\hat{n} + \cos(\theta_i)\hat{t}] \cdot \hat{n}\}\hat{n}$$

$$\hat{e}_r = -e_{i\theta'}[\sin(\theta_i)\hat{n} + \cos(\theta_i)\hat{t}] + 2e_{i\theta'}\{\sin(\theta_i)\}\hat{n}$$

$$\hat{e}_r = e_{i\theta'}[\sin(\theta_i)\hat{n} - \cos(\theta_i)\hat{t}]$$

$$\hat{e}_r = e_{i\theta'}[-\hat{\rho}']$$

$$e_{r\rho'} = -e_{i\theta'}$$



Summarizing:

Polarization of the reflected field: $\hat{e}_r = e_{r\phi} \hat{\phi}' + e_{r\rho} \hat{\rho}'$

$$e_{r\rho'} = -e_{i\theta'}$$

$$\hat{e}_r = e_{r\phi'}\hat{\phi}' + e_{r\rho'}\hat{\rho}' = -e_{i\phi'}\hat{\phi}' - e_{i\theta'}\hat{\rho}'$$

